# A Radio and X-ray Bound on Bondi-fed Accretion onto the Candidate Intermediate-Mass Black Hole in Omega Centauri, with an Infrared Consistency Check

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August 2026

v2.11 (last revised 2026-11-02) — sixth paper of the set (Paper F); companion to the Macro Transcension Hypothesis (Paper A), the inward-migration review (Paper B), the Omega Centauri campaign (Paper C), the migration economics (Paper D), the engineered-IMBH systems paper (Paper E), the residual X-ray census (Paper G),
and the mass-tension paper (Paper H);
prepared for omegacentauri.me

**Abstract**

Three deep non-detections constrain accretion onto the candidate intermediate-mass black hole (IMBH) in Omega Centauri: 170 hours of ATCA radio imaging at 1.1 $\mu$Jy rms (Mahida et al., 2026), JWST imaging with no accretion-like point source (Chen et al., 2026), and a 291-ks Chandra limit $L_X \leq 1.6 \times 10^{30}\,\mathrm{erg\,s^{-1}}$ (Haggard et al., 2013). We build a forward model from the Bondi radiative efficiency $\varepsilon_\mathrm{B} \equiv L_\mathrm{bol}/(\dot{M}_\mathrm{B} c^2)$, the mass and eight shared nuisance parameters, and report the 95 per cent credible upper limit $\varepsilon_{95}$ for three accretion-flow families. For a radiatively inefficient flow, $\varepsilon_{95} = 3.0 \times 10^{-10}$ at 8,200 $M_\odot$ and $1.1 \times 10^{-11}$ at $4 \times 10^4\,M_\odot$. These limits are conditional on the fundamental plane; without the radio leg they relax to $1.4 \times 10^{-8}$ and $8.1 \times 10^{-10}$. The infrared leg never binds, so this is a radio-plus-X-ray bound with an infrared consistency check. Against a natural outflow-suppressed hot flow at a 47 Tucanae-like gas density, the data reject 85, 77 and 99 per cent of the parameter space at 8,200 $M_\odot$, the pulsar-timing cap of 6,000 $M_\odot$ and $4 \times 10^4\,M_\odot$ (28, 20 and 76 per cent without radio). At one tenth of that density, just below the 0.03–0.08 $\mathrm{cm^{-3}}$ wind-fed estimate, the figures fall to 57, 47 and 94 per cent (4, 2 and 32). The verdict applies the lowest branch of an efficiency fit three to four decades below that branch's upper edge, and a pulsar dispersion-measure determination of the density is foreground-limited until the sight-line scatter is modelled down. Block-by-block analysis of the radio campaign excludes 100 $\mu$Jy, 2-hr flares recurring faster than every 31 hr, and leaves 1 mJy half-hour flares unconstrained beyond 3.4 days.

**Keywords:** intermediate-mass black holes; Omega Centauri; NGC 5139; Bondi accretion; radiatively inefficient accretion flows; fundamental plane of black hole activity

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# 1 Introduction

Omega Centauri hosts the nearest strong IMBH candidate: fast-moving stars within the central three arcseconds require an enclosed dark mass $\gtrsim$ 8,200 $M_\odot$ (Häberle et al., 2024), while pulsar-timing and kinematic modeling favor an extended dark remnant component of $\sim$ 2–3 $\times 10^5$ $M_\odot$ and cap any point mass at $\lesssim$ 6,000 $M_\odot$ at 3$\sigma$ (Bañares-Hernández et al., 2025), a tension sharpened rather than resolved by the current 19-pulsar timing set (Colom i Bernadich et al., 2026). The center is electromagnetically silent at the depths quantified below. The radio silence has a two-decade history: upper limits of $\sim$ 100 $\mu$Jy at GHz frequencies (Maccarone et al., 2005), a 2.5$\sigma$ near-center excursion flagged as worth deeper follow-up (Lu and Kong, 2011), JVLA limits of 1.5–2.1 $\mu$Jy beam$^{-1}$ in three other clusters that set the methodological template for radio IMBH searches (Strader et al., 2012), the 50-cluster MAVERIC survey null (Tremou et al., 2018), and now the deepest radio image of any globular cluster, 170 hours with ATCA reaching 1.1 $\mu$Jy rms at 7.25 GHz with no source at any proposed center and the nearest 5$\sigma$ source 13 arcseconds away (Mahida et al., 2026). The Lu and Kong (2011) excursion does not recur at a factor $\sim$6 deeper in rms; the 2.5$\sigma$ peak they flagged would have registered at $\gtrsim$ 14$\sigma$ in the new image, which retires that thread. In the infrared, JWST NIRCam and MIRI imaging of the core shows no point source with an accretion-like spectral energy distribution (Chen et al., 2026); in X-rays, the combined 291-ks Chandra exposure gives $L_X \leq 1.6 \times 10^{30}$ erg s$^{-1}$, which Haggard et al. (2013) translated to accretion efficiencies spanning $10^{-5}$–$10^{-9}$ across the $3 \times 10^3$–$3 \times 10^4$ $M_\odot$ mass window they consider. Their abstract rounds that span to $10^{-6}$–$10^{-8}$; the wider figure is the one their text states.

Chen et al. (2026) place their infrared limits alongside the radio limits and note that the radio is the more restrictive at the high-mass end (their Figure 7). That comparison is qualitative: each band's limit is quoted under its own model assumptions, at its own fiducial mass, with no shared error budget. The three limits constrain three different quantities, a radio flux routed through a jet or fundamental-plane relation, a band-limited infrared luminosity, and a band-limited X-ray luminosity, and all three depend on the same unmeasured gas density, the same contested black-hole mass, and the same distance. Multiplying or stacking them as if independent would overstate the joint constraint; quoting any of them at a single mass begs the question the mass tension leaves open.

This paper supplies the missing joint analysis, and in doing so obtains a result the single-band treatments could not state: the combined data are now deep enough to exclude most of the parameter space of a natural Bondi-fed hot flow across the contested mass range. Section 2 fixes the observational inputs; Section 3 the forward model, the likelihoods, and the three accretion-flow families; Section 4 the stellar-wind estimate of the dominant nuisance; Section 5 the posterior curves, the natural-flow comparison, and the prior-sensitivity budget; Section 6 the duty-cycle loophole; Section 7 forecasts; and Section 8 what the bound adjudicates. Appendix A collects every unit conversion and convention; Appendix B retains a simpler threshold-inversion construction as a transparent cross-check, and Appendix C records the version history. The Monte Carlo script, its fixed seed, and its full output accompany the paper.

# 2 Observational inputs

**Radio.** Mahida et al. (2026): ATCA project CX556 plus archival data, $\sim$170 h at 5.5 and 9.0 GHz combining to an effective 7.25 GHz image with rms $\sigma_S$ = 1.1 $\mu$Jy beam$^{-1}$; no detection at any of their three tested centers, A, B, and C, coincident with though not labelled as the van der Marel and Anderson (2010) and Noyola et al. (2008, 2010) centers. We treat the central flux density as measured to be $0 \pm \sigma_S$.

**Infrared.** Chen et al. (2026): JWST NIRCam and MIRI imaging of the core with no accretion-like point source. Their Table 1 states a limiting luminosity per filter at each of three completeness levels, and

**Table 1:** JWST per-filter limits adopted for the infrared leg, from Chen et al. (2026), their Table 1, at the 95 per cent completeness row, at their adopted distance of 5.49 kpc. $L_{\rm lim}$ is band-integrated, as that table's own definition $L = 4\pi d^2 F_\lambda \Delta\lambda$ states. $\Delta\lambda$ is not printed in the source and is recovered from the ratio $L/F_\lambda$ across that filter's completeness rows, which agree to 1–6 per cent. The likelihood predicts a band-peak $\nu L_\nu$, so the conversion between the two enters each filter's term: $\sigma_j = (\lambda_c/\Delta\lambda)\, L_{{\rm lim},j}/1.645$ (§3.3).

| Filter | Vega mag limit | $L_{\rm lim}$ (erg s$^{-1}$) | $\Delta\lambda$ (Å) | $\lambda_c/\Delta\lambda$ | $\sigma_j$ (erg s$^{-1}$) |
|---|---|---|---|---|---|
| F200W | $> 20.8$ | $4.8 \times 10^{30}$ | 4,720 | 4.24 | $1.24 \times 10^{31}$ |
| F444W | $> 17.8$ | $9.2 \times 10^{30}$ | 11,090 | 4.00 | $2.24 \times 10^{31}$ |
| F770W | $> 17.5$ | $1.9 \times 10^{30}$ | 14,985 | 5.14 | $5.94 \times 10^{30}$ |
| F1500W | $> 14.5$ | $4.9 \times 10^{30}$ | 26,850 | 5.59 | $1.66 \times 10^{31}$ |

we use those numbers directly rather than a single band-summary constant (Table 1). Each filter enters the likelihood as its own Gaussian term on the predicted band-peak $\nu L_\nu$, at the 95 per cent completeness row, the row that matches the X-ray leg's stated confidence. Their tabulated limits are band-integrated, as their own definition $L = 4\pi d^2 F_\lambda \Delta\lambda$ states, so each $\sigma_j$ carries the conversion between the two quantities as well as the quantile: $\sigma_j = (\lambda_c/\Delta\lambda)\, L_{{\rm lim},j}/1.645$, a factor of 4.0 to 5.6 per filter (Table 1). Comparing the band-peak prediction with the band-integrated limit directly would make the infrared terms tighter than the data support by that factor (Appendix C). Completeness is a detection-recovery fraction rather than a confidence level, so reading it through the normal quantile is a convention; the 99.7 and 68 per cent rows are run alongside and are reported in Appendix A. F770W dominates the combination. Treating four filters as independent terms is optimistic, since they observe one source: the conservative reading, which keeps only the tightest single filter, sits within 2 per cent of the combination on the jet family and 6 per cent on the RIAF family, so nothing here rests on the independence assumption. A second correction to the same table is disclosed rather than made silently. The 99.7 per cent completeness row for F200W is transcribed in the provenance file with a luminosity entry equal to its flux entry, which breaks the ratio $L/F_\lambda$ that the filter's other three rows share to 1 per cent; the value that ratio implies, $7.5 \times 10^{31}$ erg s$^{-1}$, is used in the 99.7 per cent ladder row of Appendix A. That row is not the primary one and no headline number depends on it. Limits are referenced at Chen et al.'s adopted 5.49 kpc and rescaled to the drawn distance. Chen et al. (2026) state that their JWST limits are more restrictive than the radio limits for $M \lesssim 6{,}000\, M_\odot$, so the $4 \times 10^4\, M_\odot$ anchor extrapolates their stated range by a factor of about seven; the SED-family treatment of Section 3.4 is what carries that anchor, and the radio-free curve there should be read with this caveat attached.

**X-ray.** Haggard et al. (2013): 290.9 ks of Chandra ACIS data, $f_X(0.5\text{–}7\,{\rm keV}) \leq 5.0 \times 10^{-16}$ erg cm$^{-2}$ s$^{-1}$ absorption-corrected. Haggard et al. (2013) quote this at 95 per cent confidence and never write a $\sigma$; their `aprates` bound is on a non-negative count rate, so we read it as one-sided and take the measured central flux as $0 \pm f_{X,{\rm lim}}/1.645$. Reading the limit as $3\sigma$, $f_{X,{\rm lim}}/3$, is not the source's convention. The two-sided 95 per cent and the $f_{X,{\rm lim}}/3$ readings are carried as a ladder in Appendix A; they span 5 per cent on the RIAF anchors, so the choice inside the ladder is not load-bearing. Su et al. (2022) re-analyse the same Chandra observations and reach a central sensitivity below $10^{30}$ erg s$^{-1}$ over 0.5–8 keV, consistent with the adopted limit; Haggard et al. (2013) stays the adopted input because band and confidence conventions differ. The deepest source catalogue of the cluster, Henleywillis et al. (2018) at 222 ks, finds no X-ray emission coincident with the centre, and any claimed central detection should be checked against it first.

**Positional coverage.** The proposed centers (van der Marel and Anderson, 2010; Noyola et al., 2008, 2010) differ by over an arcsecond, and the hole wanders about the potential minimum with r.m.s. amplitude

$\lesssim 10^{-2}$ pc, about 0.4″ at 5.49 kpc, with a leptokurtic distribution whose excursions reach 1–2″ more often than a Gaussian would allow (Di Cintio et al., 2023; Chatterjee et al., 2002). None of this strains the limits; the reason is independent of beaming: each dataset images or covers the entire core region at its quoted depth, and in the radio the nearest $5\sigma$ source of any kind lies 13″ from Center B, in Mahida et al.'s own phrasing (Mahida et al., 2026). Every position in the center list, and the full wander tail, inherits the per-band limits; no positional correction enters the error budget.

# 3 Method

## 3.1 Definitions and conventions

Let $\dot{M}_{\rm B}$ be the Bondi capture rate from gas at rest in the cluster core,

$$\dot{M}_{\rm B} = 4\pi\lambda \frac{(GM)^2 \rho_{\rm gas}}{c_s^3}, \qquad \lambda(\gamma{=}5/3) = 0.25 \text{(Bondi, 1952)}, \tag{1}$$

with $\rho_{\rm gas} = \mu_e m_p n_e$ and $\mu_e = 1.17$ for an ionized plasma of solar-like composition. Gas in hydrostatic equilibrium shares the cluster's rest frame, and the hole's Brownian velocity, $\sigma(m_*/M)^{1/2} \sim 0.1\,{\rm km\,s^{-1}}$, is negligible against $c_s$, so no bulk-velocity term enters the denominator; this is the convention of the comparison literature (Strader et al., 2012; Tremou et al., 2018; Haggard et al., 2013). A variant that treats the stellar velocity dispersion as a turbulent-velocity proxy in the denominator, the convention of Paper E's fuel budget, is evaluated in Appendix A; it lowers $\dot{M}_{\rm B}$ by a factor $\simeq 1.4$ at the median sound speed (1.9 at the 13th percentile of the prior) and weakens every limit by the same factor.

We define the *Bondi radiative efficiency*

$$\varepsilon_{\rm B} \equiv \frac{L_{\rm bol}}{\dot{M}_{\rm B} c^2}, \tag{2}$$

the bolometric radiative output per unit of captured rest-mass energy, the quantity the data constrain directly. It factors as $\varepsilon_{\rm B} = \eta\, f_{\rm B}$, with $f_{\rm B}$ the fraction of captured gas reaching the horizon and $\eta$ the radiative efficiency of the arriving flow. In natural hot flows the inflow declines inward, $\dot{M}(R) \propto R^s$, with $0 \le s \le 1$ admissible on general grounds (Blandford and Begelman, 1999) and modeled values spanning $s \simeq 0.3$–$0.8$ (Yuan and Narayan, 2014); we adopt $s \in [0.3, 0.5]$ as this paper's working sub-range within that modeled span. With $r_{\rm B} = GM/c_s^2$ and the gravitational radius $r_g = GM/c^2$, so that $r_{\rm B}/r_g = (c/c_s)^2$ and the bracket depends on $c_s$ and $s$ alone,

$$f_{\rm B} = \left(\frac{r_{\rm B}}{r_g}\right)^{-s} \in [10^{-4.4},\, 10^{-2.6}] \quad \text{for } s \in [0.3, 0.5], \tag{3}$$

consistent with Paper E's fiducial $s \simeq 0.3$ suppression of $\sim 290$ at $2 \times 10^4\, M_\odot$. Published per-band efficiencies use other conventions: Mahida et al. (2026) bound a model-specific accretion fraction ($\lesssim 4 \times 10^{-3}$), Haggard et al. (2013) an efficiency under fixed gas density and mass ($10^{-5}$–$10^{-9}$ over their stated mass window). Appendix A prints the conversion table; none of those numbers is $\varepsilon_{\rm B}$, and no direct numerical comparison should be made without it.

## 3.2 Shared nuisance parameters

Table 2 lists the priors. One draw of the full vector serves all three bands in each realization: the gas density that scales the radio prediction is the same density that scales the infrared and X-ray ones, the SED

**Table 2:** Monte Carlo priors ($4 \times 10^4$ draws, fixed seed; script `figs/fF_posterior_v4.py`). SED band fractions are family-specific (Table 3).

| Parameter | Prior | Basis |
|---|---|---|
| $n_e$ | log-normal, median 0.23 cm$^{-3}$, 0.5 dex | 47 Tuc analogy (Freire et al., 2001; Abbate et al., 2018); §4 |
| $c_s$ | uniform, 11.7–16.6 km s$^{-1}$ | $10^4$ K photoionized gas, isothermal ($\mu$=0.6) to adiabatic (pure H); the centre is the photoionized case argued for in §4, and the bracket varies the equation of state at fixed $10^4$ K without varying the temperature |
| $d$ | normal, 5.49 ± 0.05 kpc | cluster distance, centred on the set fiducial (§3.2); rescales all bands |
| $f_X$, $f_{\rm IR}$ | family-specific (Table 3) | hot-flow SED literature |
| FP latent offset | normal, 0 ± 0.88 dex | fundamental-plane intrinsic scatter (Merloni et al., 2003) |
| $s$ | uniform, 0.3–0.5 | inflow-suppression index (Yuan and Narayan, 2014); Paper E fiducial 0.3; enters $\varepsilon_{\rm nat}$ only, not the likelihood (§5.3.1) |
| $\delta$ | uniform over $\{10^{-3}, 10^{-2}, 0.1, 0.5\}$ | electron-heating fraction; the four values Xie and Yuan (2012) tabulate, drawn discretely because the efficiency fit exists only at those values; enters $\varepsilon_{\rm nat}$ only (§3.5) |
| $\sigma$ | fixed, 21 km s$^{-1}$ | paper-set fiducial (enters $r_{\rm infl}$ only) |

fraction $f_X$ that converts the X-ray measurement also drives the fundamental-plane prediction (the two legs constrain the same source's SED, so drawing $f_X$ once induces the correlation reality does), and the distance rescales all three measured quantities together.

The gas density is the weakest input and it multiplies everything, so its treatment is spread across three places: the prior here (a 47 Tucanae analogy, since $\omega$ Cen has no direct ionized-gas measurement; the one $\omega$ Cen-specific figure, $n_e = 1.94\,{\rm cm}^{-3}$ from the pulsar dispersion-measure scatter, should in its authors' words be regarded as an upper limit, since their sodium correction removes only ∼20 per cent of that scatter and, in their reading, likely underestimates a foreground responsible for much of the rest (§VI.1.3); it is not used; Wang et al. 2025), the wind budget of Section 4, which estimates the local supply, and the sensitivity budget of Section 5.3. Mahida et al. (2026) adopt a gas density of $0.2 \pm 0.1\,{\rm cm}^{-3}$ following Strader et al. (2012). In the provenance audit of Paper I this is a gas number density, the quantity Strader et al. write $\rho$ and distinct from the electron densities they measure, and Tremou et al. (2018) state that it corresponds to an electron density of $0.1\,{\rm cm}^{-3}$; labelling it $n_e$ carries a factor of two that the symbol hides. The 0.2 originates in Freire et al. (2001), who found the dispersion measures of four millisecond pulsars in M15 consistent with a homogeneous plasma of about $0.2\,{\rm cm}^{-3}$, as an alternative to the original analysis of the same pulsars (Anderson, 1993), which attributed their variation to the Galactic foreground and set only an upper limit of $5\,{\rm cm}^{-3}$ on the core electron density.[1] The value's later agreement with the 47 Tuc measurement (Abbate et al., 2018) is the analogy on which the median used here also rests, so it does not test that median independently. Every predicted flux in the forward model scales linearly with $n_e \varepsilon_{\rm B}$, so the data constrain the product, and the normalization of every curve in this paper is conditional on the $n_e$ prior.

[1] Both Freire et al. (2001) and Mahida et al. (2026) cite the thesis as Anderson (1992); the Caltech record dates it 1993.

Every part of the $\varepsilon_{\rm B}$ curve depends on the 47 Tuc analogy; what can be done, and is done below, is to bound the analogy's plausible range astrophysically and to show the answer across that range.

The distance prior is centred on the paper set's fiducial of 5.49 kpc (Appendix A.1), the value Chen et al. (2026) adopt and within 0.1 per cent of the kinematic $5.494 \pm 0.061$ kpc of Mahida et al. (2026); its width, 0.05 kpc, is that of the $5.43 \pm 0.05$ kpc of Baumgardt and Vasiliev (2021), on which versions up to 2.3 were centred. A more distant source is fainter at fixed luminosity, so every predicted flux falls and the limits loosen. Moving the centre from 5.43 to 5.49 kpc raises every $\varepsilon_{95}$ by 1.2 to 3.1 per cent (median 2.4), a mixture of the $(5.49/5.43)^2 = 1.022$ of the luminosity-derived legs and the 1.04 of the radio leg, which enters through the fundamental plane's 0.60 index as $(5.49/5.43)^{2/0.60}$, and lowers the exclusion fractions by up to 0.0033.

### 3.3 Forward model and likelihoods

For each trial $(\varepsilon_{\rm B}, M)$ and nuisance draw $\theta$: $L_{\rm bol} = \varepsilon_{\rm B}\, \dot{M}_{\rm B}(M, \theta)\, c^2$. The predicted observables are

$$f_X^{\rm pred} = f_X L_{\rm bol}/(4\pi d^2), \tag{4}$$

$$(\nu L_\nu)_{\rm IR}^{\rm pred} = f_{\rm IR} L_{\rm bol}, \tag{5}$$

$$\log L_R^{\rm pred} = 0.60\, \log\!\left(k\, f_X L_{\rm bol}\right) + 0.78\, \log(M/M_\odot) + 7.33 + \delta_{\rm FP}, \tag{6}$$

where $k = 0.610$ converts the 0.5–7 keV band fraction to the 2–10 keV band the fundamental plane is calibrated in (power law, $\Gamma = 2$; Haggard et al.'s own $\Gamma = 2.3$ gives $k = 0.462$ and is carried as a ladder row in Appendix A), and $\delta_{\rm FP} \sim \mathcal{N}(0, 0.88\ {\rm dex})$ is the plane's intrinsic scatter as a latent variable. The plane is used only in the forward direction, the direction it was fit in (Merloni et al., 2003); no inverse regression is performed, which removes the calibration-problem objection that attaches to algebraic inversion of a scattered forward relation (Plotkin et al., 2012). The likelihood is the product of six Gaussians on the measured values: central radio flux density $0 \pm 1.1\ \mu$Jy, central X-ray flux $0 \pm f_{X,\rm lim}/1.645$, and four infrared band-peak luminosities $0 \pm \sigma_j$ from Table 1, the infrared terms rescaled to the drawn distance from Chen et al.'s 5.49 kpc. Where a band's published product is a limit rather than a measured value, a Gaussian whose $\sigma$ is that limit divided by the quantile of the confidence the source itself states is the reconstruction we adopt for the two legs published as limits, and Appendix A measures what it costs; the data-availability section lists exactly those numbers as the ones that would replace it, and the machinery ingests them unchanged.

The posterior at each $M$ is computed as the plain Monte Carlo mean of the likelihood over the $4 \times 10^4$ nuisance draws, which is an unweighted average because the draws come from the prior itself and no proposal reweighting is applied, on a log-spaced $\varepsilon_{\rm B}$ grid ($10^{-14}$–$10^{-2}$, log-uniform prior), and $\varepsilon_{95}$ denotes the 95 per cent credible upper limit of the normalized posterior. Re-running the anchors under eight nuisance seeds, each with its own $\delta$ stream, puts the Monte Carlo spread of $\varepsilon_{95}$ at 1.4 to 9.6 per cent of the mean across the three anchors and the three families, widest for the RIAF family and narrowest for the radio-free curve, and the spread of $P_{\rm excl}$ at 0.13 to 0.88 percentage points. Both sit an order of magnitude under the analysis-choice ranges of Section 5.3. The seed of record sits at the low end of the $\varepsilon_{95}$ range and the high end of the $P_{\rm excl}$ range; the eight-seed minima and maxima ship in `figs/fF_v4_seedspread.json`. Upper limits from non-detections are sensitive to the prior's lower cutoff; Section 5.3 quantifies that dependence (a factor of 9 to 40 when the floor rises from $10^{-14}$ to $10^{-11}$) and every quoted $\varepsilon_{95}$ states its grid. This likelihood product, not a distribution of inverted thresholds, is what Paper E's Bayesian adjudication machinery consumes; the marginal likelihood over $\varepsilon_{\rm B}$ ships as a machine-readable table with the source.

**Table 3:** Accretion-flow families. Band fractions are log-uniform within the stated ranges, set conservatively by construction to bracket each family's characteristic SED shape rather than fit to a specific source.

| Family | $f_X$ (0.5–7 keV) | $f_{\rm IR}$ (band peak) | radio leg |
|---|---|---|---|
| RIAF (ADAF-like) | 0.03–0.3 | 0.03–0.3 | fundamental plane |
| Jet-dominated | 0.01–0.1 | 0.1–0.5 | fundamental plane |
| Thin disk | 0.05–0.2 | 0.05–0.3 | none |

### 3.4 Accretion-flow families

A single SED bracket cannot serve all flows, and under a common bracket for $f_X$ and $f_{\rm IR}$ the X-ray leg dominates the infrared leg in every draw, leaving JWST unconstraining in this band, which would make a three-band claim cosmetic. We therefore evaluate three named families with distinct band-fraction priors (Table 3): a radiatively inefficient ADAF-like flow (X-ray-bright among the bands considered), a jet-dominated, synchrotron-peaked flow (infrared-bright, X-ray-faint, the family in which the infrared leg comes closest to binding), and a thin disk (radiatively efficient; included so the reader sees by how many decades it is excluded, and evaluated without the fundamental-plane leg since the plane is a hard-state/quiescent relation).

### 3.5 The natural-flow expectation and its efficiency law

The limit curve becomes a verdict only against a prediction, and the prediction is $\varepsilon_{\rm nat} = \eta_{\rm ADAF}(\dot{m}_{\rm hor})\, f_{\rm B}$, with $\dot{m}_{\rm hor} = f_{\rm B}\dot{M}_{\rm B}/\dot{M}_{\rm Edd}$ the rate reaching the horizon in Eddington units. Xie and Yuan (2012) give $\eta$ for hot flows as a piecewise power law,

$$\eta(\dot{m}) = \eta_0 \left(\frac{\dot{m}}{10^{-2}}\right)^{a}, \tag{7}$$

with $(\eta_0, a)$ tabulated per branch and per $\delta$, the fraction of turbulent dissipation delivered directly to electrons. In the lowest branch, the one relevant here, $a$ lies between 0.59 and 0.71 (0.65, 0.59, 0.69 and 0.71 at $\delta = 0.5$, 0.1, $10^{-2}$ and $10^{-3}$; the 0.47 of the $\delta = 10^{-3}$ row belongs to the middle branch) and $\eta_0$ runs from 0.065 to 1.58 across $\delta = 10^{-3}$ to 0.5. We adopt their Table 1 as printed and carry $\delta$ as an eighth nuisance, drawn uniformly over the four tabulated values, since no measurement of it exists for this object. Like the suppression index $s$, $\delta$ enters $\varepsilon_{\rm nat}$ and not the likelihood, so its whole influence runs through the comparison rather than through $\varepsilon_{95}$.

A law linear in $\dot{m}$, $\eta = 0.1\ \min(1, \dot{m}/10^{-2})$, the form used up to v1.5, is not a simplification of these fits: at the rates this problem reaches it is lower than any tabulated $\delta$ by one to three decades, because $a = 1$ falls away far faster than $a \simeq 0.6$. Using the fits raises the expectation band by about two decades and is the larger of the two reasons the verdict of Section 5 differs from v1.5 (Appendix C).

A caveat accompanies the correction and is not removed by it. The horizon-scale rates here are $\dot{m}_{\rm hor} \simeq 10^{-9}$ to $10^{-6}$, with a median of $1.8 \times 10^{-8}$ at the fast-star anchor, while the lowest branch of Xie and Yuan's piecewise fit (their Table 1) is tabulated for $\dot{m}_{\rm net} \lesssim 1.6 \times 10^{-5}$ to $9.4 \times 10^{-5}$ depending on $\delta$, with no lower edge printed. Every number in this paper's expectation band therefore applies that branch below its upper edge, as the linear law it replaces also did: by 2.9 to 3.7 decades at the median rate ($\log_{10}$ of $1.6 \times 10^{-5}$ and $9.4 \times 10^{-5}$ over $1.8 \times 10^{-8}$), and by 1.2 to 5.0 decades across the full range $\dot{m}_{\rm hor} = 10^{-9}$ to $10^{-6}$. The direction of the extrapolation matters for how the verdict should be read: continuing the power law downward gives the smallest $\eta$ of any reading, so it is the choice least favourable to exclusion. Holding $\eta$ flat at the value on the fit's own upper edge, the opposite bracket, drives every RIAF-family entry in Table 6 to 0.99 or above, and the radio-free entries at 6,000 and 8,200 $M_\odot$ to 0.87 and 0.92. The

extrapolated law is what the headline numbers use, and the two rows below the rule in that table are the bracket.

Sgr A* offers one consistency point inside that gap. Taking a horizon accretion rate of order $10^{-8}\,M_\odot\,\mathrm{yr}^{-1}$ (Faraday-rotation limits of $2\times10^{-9}$ to $2\times10^{-7}$, Marrone et al. 2007) and a bolometric luminosity of about $10^{36}\,\mathrm{erg\,s}^{-1}$ (Yuan and Narayan, 2014), with $\dot{M}_{\rm Edd} = 10\,L_{\rm Edd}/c^2 \simeq 0.09\,M_\odot\,\mathrm{yr}^{-1}$ for $4\times10^6\,M_\odot$ (Event Horizon Telescope Collaboration, 2022), gives $\dot{m} \simeq 10^{-7}$ and $\eta = L/(\dot{M}c^2) \simeq 2\times10^{-3}$. The lowest branch of Xie and Yuan continued to $\dot{m} = 10^{-7}$ gives $9\times10^{-4}$ at $\delta = 0.5$ and $1.3\times10^{-4}$ at $\delta = 0.1$, so Sgr A* sits inside the $\dot{m}_{\rm hor}$ range of this paper, near the $\delta = 0.5$ branch. The comparison is limited in several ways. The $\delta$ values of Xie and Yuan (2012) follow models calibrated on Sgr A*-type sources, so the agreement is not independent of the fit. It fixes one $(\dot{m}, \delta)$ point, so using it assumes that $\omega$ Cen shares the $\delta$ of Sgr A*. And at the median $\dot{m}_{\rm hor}$ of $1.8\times10^{-8}$ it leaves a factor of 5.6 in $\dot{m}$ (0.75 decade) to extrapolate. Figure 1 does not mark it.

### 3.6 Defining the exclusion fraction

Each natural-flow draw is a fully specified model: $\varepsilon = \varepsilon_{\rm nat}(\theta)$ with that draw's own nuisances. It predicts a flux in every observed band, and the measured data, zero central flux at the published $\sigma_j$, then have a joint chi-square under it, $\chi^2 = \sum_b (\mathrm{pred}_b/\sigma_b)^2$ over the six terms of Section 3.3, distributed as $\chi^2_6$ if that draw is the truth. We call a draw excluded when the data reject it at the 5 per cent level, and report

$$P_{\rm excl} = P\left[\, p(\chi^2) < 0.05 \,\right] \tag{8}$$

over the nuisance draws, with its Monte Carlo standard error.

A statistic reported under the same name up to v1.5 (Appendix C) was a different quantity: the fraction of natural draws exceeding $\varepsilon_{95}$, with $\varepsilon_{95}$ itself marginalized over those same draws. That statistic compares a distribution against a scalar built from it, uses common random numbers on both sides, and has no coverage reading; it also produced the artefact that widening the density prior cost the verdict almost nothing, since a log-normal's median does not move when its width does, so the exceedance count barely changed. Equation 8 avoids all three. It is per-draw, it uses the likelihood rather than a summary of it, and it never reads $\varepsilon_{95}$, which is why the efficiency prior's lower cutoff, a factor of 9 to 40 on $\varepsilon_{95}$, cannot touch it. The old statistic is retained in Table 4 under its correct name, recomputed on independent draws so that the common random numbers are gone; it runs 0.03 to 0.04 above the posterior-predictive figure at the two lower anchors and reaches 1.00 at the high anchor where the posterior-predictive value is 0.99.

## 4 The stellar-wind gas supply

The $n_e$ prior needs an astrophysical anchor, and the cluster's own stellar winds supply one. A recount of its inputs (`figs/fF_calc7_windfloor.py`) shows that the budget is a no-removal estimate; v1.5 and earlier used it as a floor, and the correction runs to two decades (Appendix C).

The chain is the standard one. Inside the influence radius $r_{\rm infl} = GM/\sigma^2 = 0.39\,\mathrm{pc}$ of a $4\times10^4\,M_\odot$ hole, a Plummer model of the cluster ($M = 4\times10^6\,M_\odot$, $a = 3.78\,\mathrm{pc}$) encloses $4.3\times10^3\,M_\odot$ of stars, about $7.2\times10^3$ stars at a luminous-mass mean of $0.6\,M_\odot$. Of those, the Kroupa weight of the turnoff window is 0.035 and the fraction of a turnoff star's main-sequence life spent above $10^3\,L_\odot$ is $(0.35+0.07)/13 = 0.032$, so the standing giant population inside $r_{\rm infl}$ is 8.1 stars, and 5.2 to 11.0 across the phase-lifetime and mean-mass variations tested. At $10^{-8}\,M_\odot\,\mathrm{yr}^{-1}$ each the aggregate injection is $8.1\times10^{-8}\,M_\odot\,\mathrm{yr}^{-1}$, and if

nothing removes it the steady state at an assumed outflow speed $v_{\rm out} \sim 50\,{\rm km\,s^{-1}}$ is

$$n_e \sim \frac{\dot{M}_{\rm wind}}{4\pi r_{\rm infl}^2\,\mu_e m_p v_{\rm out}} \simeq 0.029\,{\rm cm^{-3}}, \tag{9}$$

with 0.018 to $0.039\,{\rm cm^{-3}}$ over the same variations, refilled on the crossing time $r_{\rm infl}/v_{\rm out} \sim 10^4$ yr. The superseded count of $10^3$ giants reproduces $3.5\,{\rm cm^{-3}}$ through the same arithmetic and was too large by a factor of 123, having counted the stars rather than the ones currently on the giant branch (Appendix C).

The recount changes three things.

*The wind budget is an estimate.* A no-removal steady state is the most gas the winds can sustain at a given outflow speed, but $0.029\,{\rm cm^{-3}}$ is not a ceiling: it is linear in the assumed $v_{\rm out} \sim 50\,{\rm km\,s^{-1}}$, and an outflow slowed to the sound speed ($v_{\rm out} \approx c_s$) raises it to about $0.10\,{\rm cm^{-3}}$. Three estimates bracket the wind-fed density at 0.03–$0.08\,{\rm cm^{-3}}$: this recount (0.029), the Haggard et al. (2013) wind budget (0.038, below), and the $0.0764\,{\rm cm^{-3}}$ upper edge of the $\omega$ Cen envelope in the gas-provenance audit of Paper I (its tier table). The retired $n_e \geq 0.05\,{\rm cm^{-3}}$ floor sits a factor 1.7 above it, and the claims made for that floor are withdrawn (Appendix C). Section 5.3 now reports a truncation at $n_e \geq 0.02\,{\rm cm^{-3}}$, inside the recounted range, and states what it costs rather than what it recovers.

*The classical gas problem does not arise here.* Globular clusters are gas-poorer than their stellar winds imply, and in 47 Tuc a removal agent, most plausibly millisecond-pulsar winds and heating, is required to explain it (Freire et al., 2001; Abbate et al., 2018). At 8 giants inside 0.39 pc there is little to remove: the wind supply and the 47 Tuc analogy value differ by a factor of 8 in the direction that needs no removal agent at all.

*The recount bears on the prior's centre.* The $0.23\,{\rm cm^{-3}}$ median is imported from 47 Tuc, and $\omega$ Cen's own giants inside the influence radius cannot supply it without a source this budget omits. An independent route reaches a similar figure: Haggard et al. (2013) adopt $n_e = 0.038\,{\rm cm^{-3}}$ for this cluster from a Pfahl–Rappaport wind budget, within 32 per cent of the recount here despite a different formalism and radius. Either the central density is set by material the influence-radius wind budget does not track, in which case the 47 Tuc analogy stands and the prior is right, or it is set by the local supply, in which case the prior median is high by a factor of 3 to 8 (0.23 against 0.0764 to 0.029) and every limit in this paper is correspondingly loose. We do not collapse that into one number. The prior is left at its published centre so that the curves remain comparable with the comparison literature, the $n_e$ median/10 row of Table 5 is the quantitative price of the second reading, and Section 7 states that a pulsar dispersion-measure gradient is the measurement that would end the argument and that the present sight-line scatter keeps it out of reach until that scatter is modelled. On the second reading the density evidence sits at the weaker end of this paper's exclusion range, which is the direction Table 6's prior-shift row prices.

## 5 Results

Figure 2 shows the full joint posterior behind these curves over the two parameters of interest and the two dominant nuisances, RIAF family only.

Figure 1 and Table 4 carry the content. Three statements summarize them.

*The radio leg dominates wherever the fundamental plane applies.* The plane's mass term rises as $M^{0.78}$ while the Bondi denominator rises as $M^2$, which sets the direction; the realized slope of the marginalized limit is shallower than that single-draw asymptote, $\varepsilon_{95} \propto M^{-2.1}$ between the two upper anchors and $M^{-2.2}$ fitted across the whole grid, because the plane's 0.88-dex scatter and the two flux legs flatten the response. The asymptote $M^{-3.3}$ quoted in v1.5 is not the realized slope. The radio-driven curve separates from the radio-free curve by 1.9 decades at the top of the mass range. This is the single-object version of the

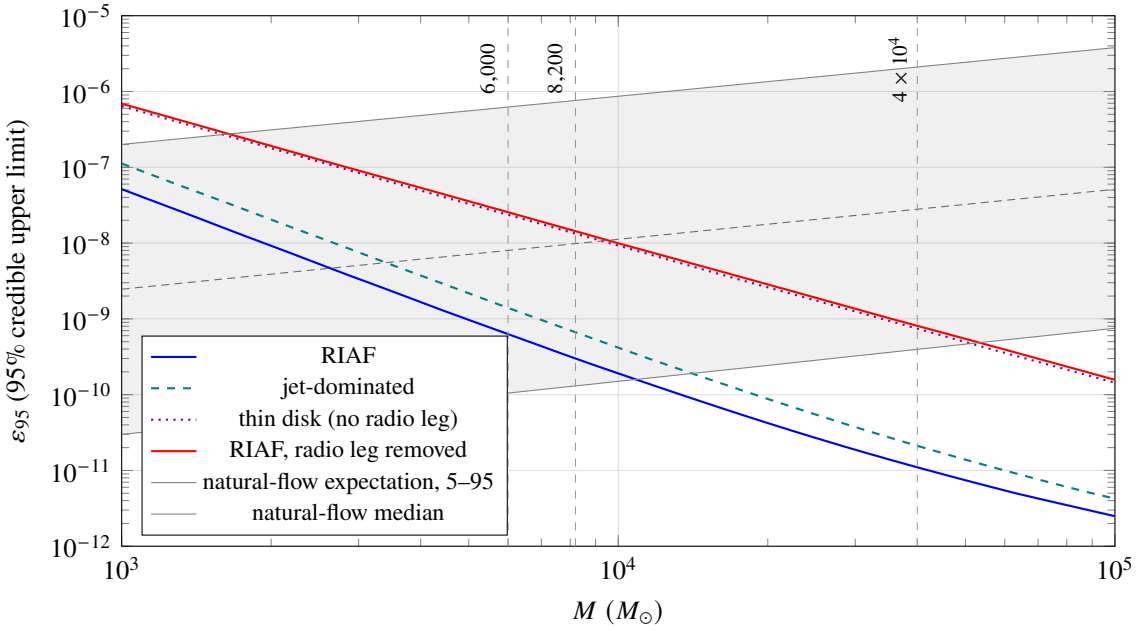



**Figure 1:** 95 per cent credible upper limits on the Bondi radiative efficiency $\varepsilon_{\rm B}$ of any central accretion flow in $\omega$ Cen, per accretion-flow family, with the expectation band for a natural outflow-suppressed hot flow (grey: 5–95 percentile of $\varepsilon_{\rm nat} = \eta_{\rm ADAF}(\dot{m})\, f_{\rm B}$ over the same nuisance draws, on the Xie and Yuan (2012) efficiency of §3.5) rising while the limits fall. The band median sits above the RIAF limit from about $2.6 \times 10^3\ M_\odot$ upward, below the contested masses, so it does not cross the limit inside them (the band's 5th-percentile edge stays below the limit to about $1.1 \times 10^4\ M_\odot$); this is why the exclusion fractions of Table 4 are high at every anchor; the vertical separation is not itself the exclusion fraction, which §3.6 defines from the likelihood rather than from this figure. Vertical lines mark the pulsar-timing point-mass cap (Bañares-Hernández et al., 2025), the fast-star lower bound (Häberle et al., 2024), and the upper kinematic range. All curves conditional on the $n_e$ prior of Table 2.

**Table 4:** $\varepsilon_{95}$ at the three mass anchors (evaluated at the anchor masses, not grid neighbours), and the posterior-predictive exclusion fraction $P_{\rm excl}$ of §3.6: the fraction of natural-flow draws whose own predicted observables the measured data reject at the 5 per cent level. Monte Carlo standard errors on $P_{\rm excl}$ are below 0.003 throughout. The last column is the prior-exceedance fraction that v1.5 reported under the name $P_{\rm excl}$, recomputed here on independent draws.

| $M$ ($M_\odot$) | RIAF | jet | thin disk | RIAF, no radio | $P_{\rm excl}$ (RIAF) | prior exc. (RIAF) |
|---|---|---|---|---|---|---|
| 6,000 | $6.2 \times 10^{-10}$ | $1.4 \times 10^{-9}$ | $2.4 \times 10^{-8}$ | $2.5 \times 10^{-8}$ | 0.77 | 0.82 |
| 8,200 | $3.0 \times 10^{-10}$ | $6.6 \times 10^{-10}$ | $1.3 \times 10^{-8}$ | $1.4 \times 10^{-8}$ | 0.85 | 0.90 |
| $4 \times 10^4$ | $1.1 \times 10^{-11}$ | $2.1 \times 10^{-11}$ | $7.4 \times 10^{-10}$ | $8.1 \times 10^{-10}$ | 0.99 | 1.00 |

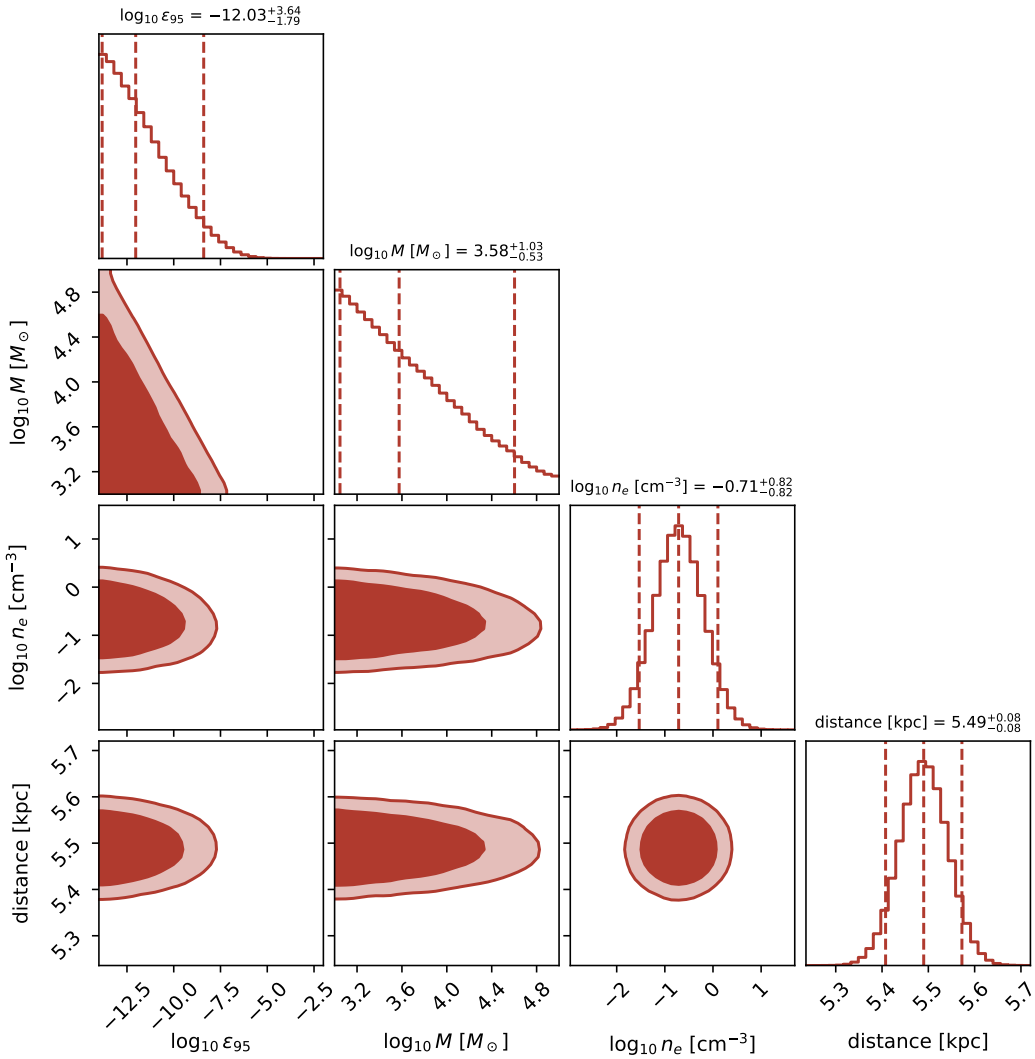



**Figure 2:** Joint posterior corner plot for the RIAF family over the two parameters of interest ($\log_{10}\varepsilon_{95}$, the Bondi efficiency whose 95 per cent credible upper limit Figure 1 reports, and $\log_{10} M$) and the two dominant nuisances ($\log_{10} n_e$ and the distance). Samples from the shipped Monte Carlo (same seed, priors, and likelihood; $\varepsilon_{\rm B}$ log-uniform over the efficiency grid, per-sample weights $e^{\ln L}$; effective sample size $1.06 \times 10^5$ of $4 \times 10^5$ draws; script `figs/fF_corner_plot_v2.py`). This figure alone adds a prior on $M$, log-uniform over the mass grid's range, because a corner plot needs a joint sample and the rest of the paper deliberately reports $\varepsilon_{95}$ at fixed $M$ instead of marginalizing over a mass prior the kinematic literature does not agree on. The $M$ panel therefore shows what that particular choice implies and is not a result of the paper. Contours enclose 68 and 90 per cent of the posterior weight. The $\varepsilon_{\rm B}$–$M$ degeneracy is the flux-limit constraint: the predicted X-ray, infrared, and radio luminosities all scale as $\varepsilon_{\rm B}\, M^2$ (Bondi) or steeper, so the posterior's upper envelope in $\varepsilon_{\rm B}$ falls as $M$ rises, the same structure Figure 1 displays curve by curve. The distance posterior reproduces its prior: the limits are flux measurements, and the drawn distance enters the prediction rather than the limit.

population-level result of Maccarone et al. (2005), Strader et al. (2012), and Tremou et al. (2018): for IMBH masses, radio provides the tightest constraint on quiescent accretion. The plane's extrapolation into deep quiescence is contested (Plotkin et al., 2012), its 0.88-dex scatter is carried as a latent variable rather than a point estimate, and readers who distrust it entirely should quote the red curve: $\varepsilon_{95} = 1.4 \times 10^{-8}$ at 8,200 $M_\odot$ and $8.1 \times 10^{-10}$ at $4 \times 10^4$ $M_\odot$, from Chandra and JWST with no shared emission model.

*Against the natural-flow expectation, the verdict holds across the contested mass range.* The grey band is the prediction, $\varepsilon_{\rm nat} = \eta_{\rm ADAF}(\dot{m})\, f_{\rm B}$, evaluated draw by draw on the same nuisances, with $\eta_{\rm ADAF}$ now the piecewise fit of Xie and Yuan (2012) as summarized in Yuan and Narayan (2014) and $\delta$ carried as a nuisance over the four values they tabulate (§3.5). This expectation carries no $\delta_{\rm FP}$ term, unlike $\varepsilon_{95}$ for the RIAF and jet families, whose radio-leg prediction does (Section 3.3); the fundamental-plane scatter widens the limit but not the band it is compared against, so a reader who distrusts the plane should read the exclusion fractions off the radio-free column rather than treating the asymmetry as a wash. The band rises with mass while every limit falls, and it now sits above the RIAF limit across the whole contested range rather than crossing inside it. At the pulsar-timing cap and the fast-star anchor the data reject 77 and 85 per cent of natural-flow draws under the RIAF family, and 20 and 28 per cent with the radio leg removed; at $4 \times 10^4$ $M_\odot$ the figures are 99 and 76 per cent. Version 1.5 reported 19, 31 and 90 per cent for the RIAF column; the two corrections listed in Appendix C both move the verdict the same way. Table 6 carries the figures across the sensitivity ladder. A Bondi-fed hot flow at the fiducial density is now disfavoured across the contested mass range and not only at its top, which is a stronger claim than that of v1.5 and a more fragile one, since it rests on applying the lowest branch of a published efficiency fit well below its upper edge. Section 3.5 states what that extrapolation is worth and Table 6's last two rows bracket it.

*The thin disk is excluded everywhere.* $\varepsilon_{95}$ for the disk family sits six to nine decades below the $\eta \sim 0.06$–$0.4$ a disk radiates at; no Bondi-fed thin disk at any mass in the range survives, quantifying what every prior single-band paper assumed informally.

### 5.1 The exclusion fraction as a continuous function of mass

Table 4 evaluates $P_{\rm excl}$ at three masses because those are the masses the kinematic literature argues about. The Monte Carlo evaluates it on the whole grid, and the continuous version (Figure 3) is the more useful object for a reader who holds a different mass prior than any of the three anchors. It also locates the transitions. Under the RIAF family the exclusion fraction passes 50 per cent at $2.5 \times 10^3$ $M_\odot$ and 90 per cent at $1.1 \times 10^4$ $M_\odot$; under the jet family the same thresholds fall at $3.3 \times 10^3$ and $1.4 \times 10^4$ $M_\odot$. Using no radio information they move to $1.7 \times 10^4$ and $7.9 \times 10^4$ $M_\odot$. The 50 per cent threshold sits below the pulsar-timing point-mass cap of 6,000 $M_\odot$ and the 90 per cent one, at $1.1 \times 10^4$ $M_\odot$, above it, so on the efficiency law adopted here the constraint bites across the contested range and not only at its top.

### 5.2 The binding band in each family

Figure 4 shows what the family definitions of Section 3.4 change. At each anchor the figure evaluates the forward model at that family's own $\varepsilon_{95}$ and plots the predicted $\nu L_\nu$ in every observed band against the published limit in that band, so the three instruments and the three families appear on one axis.

The radio leg binds every family that carries it, and by a wide margin. For the RIAF family the median predicted radio flux density at $\varepsilon_{95}$ sits at 1.5 times the ATCA $5\sigma$ point at 8,200 $M_\odot$ and 4.8 times it at $4 \times 10^4$ $M_\odot$, while the X-ray prediction sits at about 1 per cent of the Chandra bound and the tightest infrared filter at 0.2 per cent of its own. For the jet family, the one built to be infrared-bright, the radio ratios are 1.2 and 3.5 and the F770W ratio reaches 0.8 and 0.6 per cent: better than under the RIAF band

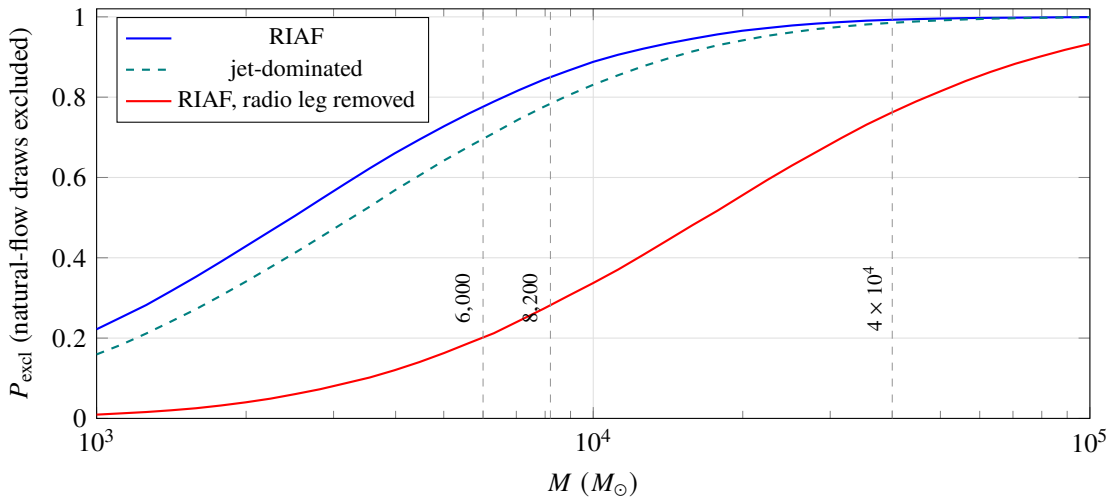



**Figure 3:** Fraction of natural-flow parameter space excluded, $P_{\rm excl}$ of Eq. 8, as a continuous function of mass: every natural-flow draw scored against the measured data at every grid mass rather than at three. Vertical lines mark the pulsar-timing point-mass cap (Bañares-Hernández et al., 2025), the fast-star lower bound (Häberle et al., 2024), and the upper kinematic range. The three anchor values of Table 4 sit on these curves by construction. Coordinates from `figs/fF_expand_exclcurve_v2.json`. The quantity plotted is the posterior-predictive fraction of §3.6; that file also carries the prior-exceedance curve the v1.5 figure plotted, so the change is auditable.

fractions by a factor of five, and still short of the plane by more than two decades. JWST does not bind for this family, and the band-integration conversion of Section 2 widens the gap.

The thin disk carries no radio leg, and there the X-ray band binds alone, at 0.44 and 0.59 of the Chandra bound against F770W's 0.09 and 0.12. With the prediction and the limit expressed as the same quantity, band-integrated luminosity, the infrared sits a factor of five to six below the X-ray in the RIAF and thin-disk families, and a factor of 1.2 above it in the jet family. In summary this is a radio-plus-X-ray bound with an infrared consistency check, not a three-way joint constraint, and Section 7 prices deeper mid-infrared accordingly.

Ratios above unity for the radio band are expected and are not a detection claim: $\varepsilon_{95}$ is a 95 per cent posterior quantile marginalized over the plane's 0.88-dex latent scatter, so at that $\varepsilon_{\rm B}$ the median draw predicts a flux the image would have seen while a substantial tail does not. The five-percentile whiskers, which reach two decades below the medians in the radio and a factor of ten in the other bands, are that scatter. The infrared points are not flat across the four filters: each is converted to that filter's band-integrated luminosity by its own $\Delta\lambda/\lambda_c$ (Table 1), which is the quantity the published limits are. F770W still sets the infrared constraint, by a factor of two over the next filter, because it combines the tightest limit with the narrowest fractional bandwidth.

## 5.3 Sensitivity budget

Table 6 prices the headline verdict rather than the limit behind it, and the two do not move together. Widening the $n_e$ prior costs the RIAF exclusion fraction 0.10 at the fast-star anchor and 0.05 at the high anchor; shifting the prior centre down a decade costs 0.27 at the fast-star anchor while leaving the high anchor at 0.94, because at $4 \times 10^4\ M_\odot$ the natural-flow band clears the limit by enough margin that a factor of ten in density does not close it. The prior-shift row is still the widest single move, and §4's recount is an argument that it is the more likely direction rather than a worst case. The two rows at prior medians

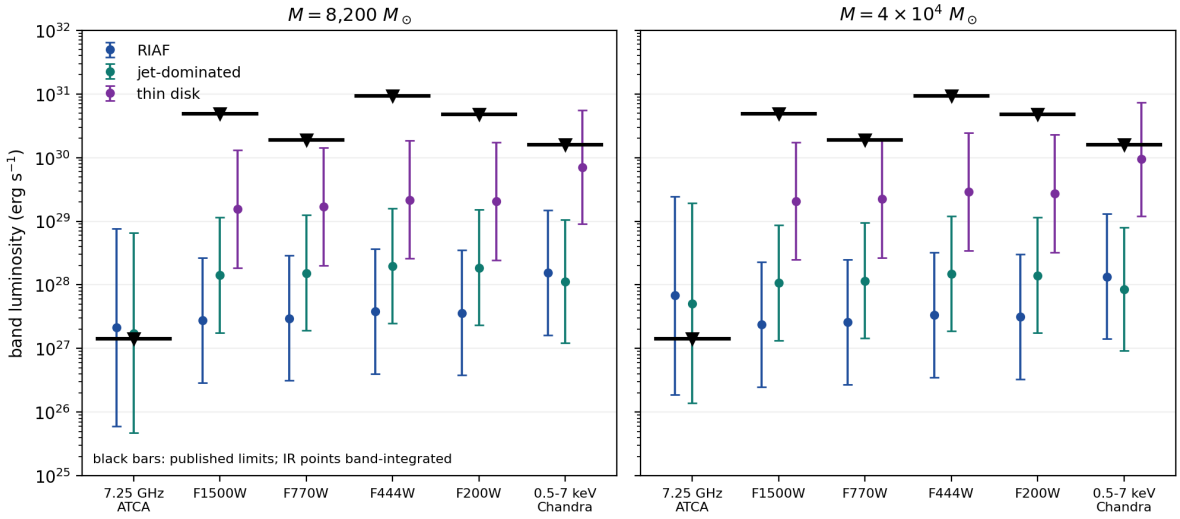



**Figure 4:** Predicted band luminosities at each family's own $\varepsilon_{95}$, against the published limits (black bars, arrows marking the excluded direction). Points are medians over the $4 \times 10^4$ nuisance draws, whiskers the 5th to 95th percentiles. The thin-disk family has no radio point because the fundamental-plane leg is not applied to it (Section 3.4). The radio limit is drawn at $5\sigma$ (5.5 $\mu$Jy) so that it is comparable with the X-ray and infrared bounds, which are published as limits; the likelihood itself uses the 1.1 $\mu$Jy rms as a Gaussian $\sigma$ and is unaffected. Infrared points are band-integrated luminosities, the same quantity as the published limits (§2). Script `figs/fF_expand_v2.py`, numbers in `figs/fF_expand_sed_v2.json`.

**Table 5:** Sensitivity of the RIAF-family $\varepsilon_{95}$ to the dominant analysis choices. Each row changes one ingredient from the baseline.

| Variant | 6,000 $M_\odot$ | 8,200 $M_\odot$ | $4 \times 10^4\ M_\odot$ |
|---|---|---|---|
| baseline | $6.2 \times 10^{-10}$ | $3.0 \times 10^{-10}$ | $1.1 \times 10^{-11}$ |
| $n_e$ prior widened to 1.5 dex | $3.6 \times 10^{-9}$ | $1.8 \times 10^{-9}$ | $7.6 \times 10^{-11}$ |
| $n_e$ prior median $\div$ 10 | $3.8 \times 10^{-9}$ | $1.8 \times 10^{-9}$ | $4.5 \times 10^{-11}$ |
| widened + wind truncation ($n_e \geq 0.02$) | $1.1 \times 10^{-9}$ | $5.2 \times 10^{-10}$ | $1.9 \times 10^{-11}$ |
| widened + retired floor ($n_e \geq 0.05$) | $6.9 \times 10^{-10}$ | $3.4 \times 10^{-10}$ | $1.3 \times 10^{-11}$ |
| $\varepsilon_B$-grid floor raised to $10^{-11}$ | $5.6 \times 10^{-9}$ | $3.4 \times 10^{-9}$ | $4.4 \times 10^{-10}$ |
| IR band correction removed (the v1.5 leg) | $5.8 \times 10^{-10}$ | $2.8 \times 10^{-10}$ | $1.1 \times 10^{-11}$ |
| $k$ at $\Gamma = 2.3$, the source's own index | $7.5 \times 10^{-10}$ | $3.6 \times 10^{-10}$ | $1.3 \times 10^{-11}$ |
| $c_s = 11.7\,\mathrm{km\,s^{-1}}$, bracket low end | $4.0 \times 10^{-10}$ | $2.0 \times 10^{-10}$ | $7.9 \times 10^{-12}$ |
| $c_s = 16.6\,\mathrm{km\,s^{-1}}$, bracket high end | $9.0 \times 10^{-10}$ | $4.3 \times 10^{-10}$ | $1.4 \times 10^{-11}$ |
| $c_s = 37.1\,\mathrm{km\,s^{-1}}$ ($10^5$ K, isothermal $\mu$=0.6) | $6.1 \times 10^{-9}$ | $2.8 \times 10^{-9}$ | $6.4 \times 10^{-11}$ |
| Paper E's Bondi convention (its Eq. 2), $\mu_e$ unchanged | $8.2 \times 10^{-10}$ | $3.9 \times 10^{-10}$ | $1.3 \times 10^{-11}$ |

**Table 6:** The posterior-predictive exclusion fraction under the same variants, for the RIAF family and for the radio-free curve. Table 5 moves only $\varepsilon_{95}$; here the natural-flow draws move with the nuisances, since both are built from one set of draws. The $\varepsilon_{\rm B}$-grid floor has no row: the statistic of §3.6 never reads $\varepsilon_{95}$, so the efficiency prior's lower cutoff cannot reach it, which is one practical gain of the reconstruction. Monte Carlo standard errors are below 0.003 throughout.

| | RIAF | | | RIAF, no radio | | |
|---|---|---|---|---|---|---|
| Variant | 6,000 | 8,200 | $4\times10^4$ | 6,000 | 8,200 | $4\times10^4$ |
| baseline | 0.77 | 0.85 | 0.99 | 0.20 | 0.28 | 0.76 |
| $n_e$ prior widened to 1.5 dex | 0.69 | 0.75 | 0.94 | 0.33 | 0.38 | 0.64 |
| $n_e$ prior median 0.08 $\mathrm{cm}^{-3}$ | 0.64 | 0.73 | 0.98 | 0.08 | 0.13 | 0.56 |
| $n_e$ prior median 0.03 $\mathrm{cm}^{-3}$ | 0.50 | 0.61 | 0.95 | 0.03 | 0.05 | 0.37 |
| $n_e$ prior median ÷ 10 | 0.47 | 0.57 | 0.94 | 0.02 | 0.04 | 0.32 |
| widened + wind truncation ($n_e \geq 0.02$) | 0.74 | 0.81 | 0.98 | 0.33 | 0.38 | 0.69 |
| widened + retired floor ($n_e \geq 0.05$) | 0.78 | 0.85 | 0.99 | 0.34 | 0.39 | 0.75 |
| X-ray limit read two-sided 95 per cent | 0.78 | 0.85 | 0.99 | 0.21 | 0.30 | 0.78 |
| X-ray limit read as $3\sigma$ | 0.78 | 0.85 | 0.99 | 0.25 | 0.34 | 0.81 |
| $k$ at $\Gamma = 2.3$ | 0.76 | 0.83 | 0.99 | 0.20 | 0.28 | 0.76 |
| IR completeness row 99.7 per cent | 0.77 | 0.85 | 0.99 | 0.20 | 0.28 | 0.76 |
| IR completeness row 68 per cent | 0.77 | 0.85 | 0.99 | 0.21 | 0.29 | 0.77 |
| IR tightest single filter only | 0.80 | 0.87 | 0.99 | 0.23 | 0.31 | 0.79 |
| IR band correction removed (the v1.5 leg) | 0.78 | 0.85 | 0.99 | 0.24 | 0.33 | 0.80 |
| Bondi denominator, Paper E convention | 0.73 | 0.81 | 0.99 | 0.15 | 0.22 | 0.70 |
| $c_s = 11.7\,\mathrm{km\,s}^{-1}$, bracket low end | 0.82 | 0.88 | 1.00 | 0.26 | 0.35 | 0.82 |
| $c_s = 16.6\,\mathrm{km\,s}^{-1}$, bracket high end | 0.73 | 0.82 | 0.99 | 0.15 | 0.22 | 0.70 |
| $c_s = 37.1\,\mathrm{km\,s}^{-1}$ ($10^5$ K) | 0.49 | 0.60 | 0.95 | 0.02 | 0.04 | 0.35 |
| $\eta$ held at the XY12 fit's own upper edge | 0.99 | 1.00 | 1.00 | 0.87 | 0.92 | 1.00 |
| $\eta$ linear in $\dot{m}$ (the v1.5 form) | 0.36 | 0.47 | 0.92 | 0.01 | 0.02 | 0.26 |

of 0.08 and 0.03 $\mathrm{cm}^{-3}$ span the wind-recount bracket of §4: the RIAF fraction at the fast-star anchor is 0.73 and 0.61 there, and 0.13 and 0.05 with the radio leg removed. Across every convention choice in this paper the RIAF exclusion fraction at $4 \times 10^4\,M_\odot$ stays above 0.93 and at 8,200 $M_\odot$ above 0.57; the radio-free column is the fragile one, spanning 0.04 to 0.39 at the fast-star anchor and 0.32 to 0.82 at the high anchor, and it should be read with that range attached. Two rows below the rule price the efficiency prescription itself rather than a convention: holding $\eta$ at the fit's own upper edge (the top of its lowest branch), instead of extrapolating the fit 2.9 to 3.7 decades below it at the median rate, raises every entry towards unity, while the linear $\eta$ of v1.5 lowers them to the values reported there. The extrapolation this paper adopts is the weaker of the two readings of the cited fit, and the verdict is stated on that reading.

Table 5 is the quantified version of Section 3.2's caveat. Widening the $n_e$ prior by a decade in each direction, or dividing its median by ten, each relax $\varepsilon_{95}$ by a factor of about six: the curve's normalization belongs to the gas-density assumption, in both its width and its centre, and every claim in this paper carries that conditioning. Truncating the widened prior at $n_e \geq 0.02\,\mathrm{cm}^{-3}$, inside the range the wind recount of Section 4 supports, recovers about two thirds of the widening in the log: $\varepsilon_{95}$ at the fast-star anchor moves from $1.8 \times 10^{-9}$ back to $5.2 \times 10^{-10}$ against a baseline of $3.0 \times 10^{-10}$. The $\varepsilon_{\mathrm{B}}$-prior floor matters as it always does for non-detections, by factors of 9 and 11 at the two lower anchors and 40 at $4 \times 10^4\,M_\odot$ when the grid floor rises three decades, so a single factor of about ten holds only at the lower anchors. Every $\varepsilon_{95}$ here is quoted on the $10^{-14}$ grid, the machine-readable likelihood tables let any reader impose their own, and the exclusion fractions of Table 6 are now independent of that choice entirely.

### 5.3.1 Nuisance contributions to the spread

Table 5 varies analysis choices. The nuisance parameters can be interrogated separately, by re-running the same posterior with one of them held at its prior's central value while the others are drawn, and again with only that one drawn and the others held (Figure 5). The two views answer different questions: the first asks how much of the limit's looseness a given parameter is responsible for, the second asks how much looseness that parameter would produce on its own. Seven parameters appear; the eighth, $\delta$, is absent for the same structural reason as $s$, since it enters $\varepsilon_{\mathrm{nat}}$ and not the likelihood, so freezing it cannot move $\varepsilon_{95}$.

The answer is that the fundamental plane's latent scatter carries the spread. Freezing $\delta_{\mathrm{FP}}$ at zero tightens $\varepsilon_{95}$ by a factor of 13 at 8,200 $M_\odot$ and 29 at $4 \times 10^4\,M_\odot$; drawing it alone, with everything else fixed, reproduces a limit 18 and 61 times looser than the fully frozen case. No other parameter comes within an order of magnitude. The gas density is second and is a distant second in this accounting: freezing it tightens $\varepsilon_{95}$ by 21 to 25 per cent, and drawing it alone loosens the frozen limit by a factor 1.6 to 2.4. The X-ray band fraction contributes 8 to 10 per cent, the scatter of the sound speed under 1 per cent, and the distance nothing measurable, as expected from a 1 per cent prior. Freezing a nuisance measures its share of the spread, not the effect of moving its centre, and the centre of the sound-speed prior does move the limit: the bracket's own ends span a factor of 2.2 in $\varepsilon_{95}$ at 6,000 and 8,200 $M_\odot$ and 1.8 at $4 \times 10^4\,M_\odot$, and a $10^5$ K medium raises it by factors of 9.7, 9.2 and 5.9 (Tables 5 and 6; `figs/fF_cs_sensitivity.py`). The infrared band fraction moves the RIAF limit by 0.4 per cent in the opposite direction, which is the signature of a leg that never binds in this family.

The distance prior and the suppression index are inert here, for structural reasons worth naming. The distance prior is narrow enough that its 1 per cent width cannot register against the plane's 0.88 dex. The suppression index $s$ does not enter the likelihood at all: it sets the natural-flow expectation $\varepsilon_{\mathrm{nat}}$ against which the limit is compared, not the limit itself, so its entire influence in this paper runs through $P_{\mathrm{excl}}$ and the grey band of Figure 1 rather than through $\varepsilon_{95}$.

This is consistent with Section 3.2's statement that the gas density sets the normalization, because the two statements concern different things. Within the stated priors, the plane's scatter is what widens the

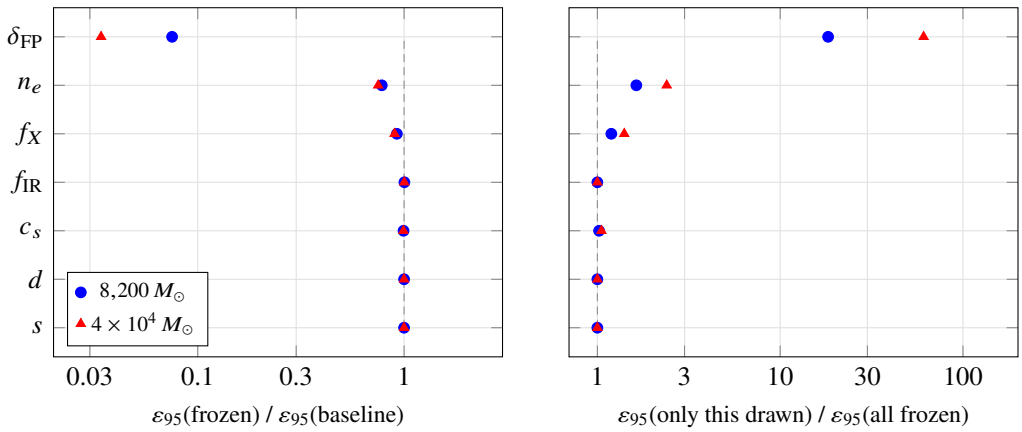



**Figure 5:** Per-nuisance contribution to the RIAF-family limit, at two mass anchors. Left: $\varepsilon_{95}$ with that nuisance held at its prior's central value, divided by the baseline $\varepsilon_{95}$; a point far to the left means that parameter's spread was inflating the limit by that factor. Right: $\varepsilon_{95}$ with only that nuisance drawn, divided by the limit with all seven frozen. Neither panel is a variance decomposition, since $\varepsilon_{95}$ is a posterior quantile rather than a sum of independent terms, and the two panels do not multiply back to unity. The distance and the suppression index sit on the reference line in both panels; $s$ does so because it enters the natural-flow expectation and not the likelihood. Numbers in `figs/fF_expand_nuisance_v2.json`.

posterior; the density's risk lies in the choice of prior centre, which no freezing test can expose, and which Table 5's prior-shift rows price at a factor ~6. A reader who distrusts the plane has the radio-free curve; a reader who distrusts the density analogy has the shift rows and the wind recount of Section 4, which points the same way.

## 5.4 Comparison with the survey population

The single-cluster depth used here is best read against the 50-cluster MAVERIC survey (Tremou et al., 2018), which is the reference population for radio IMBH limits and the source of the field's null (Table 7). The comparison needs care. First, each entry assumes its own distance: Tremou et al. (2018) adopt 4.9 kpc for $\omega$ Cen, Haggard et al. (2013) 5.2 kpc, Chen et al. (2026) and Mahida et al. (2026) 5.49 kpc, the distance this paper adopts. The spread is ordinary paper-to-paper variation and it propagates as $d^2$ into any luminosity, so the flux densities compare directly while the derived quantities do not. Second, the survey's mass limits are obtained by inverting the fundamental plane, the step Section 3.3 declines to take; they are quoted here as the literature's own currency and are not $\varepsilon_B$ and not convertible to it without the full nuisance treatment. Abbassi et al. (2026) read the $\omega$ Cen non-detection the same way: it constrains gas supply and accretion efficiency, and in their words is "not a dynamical veto" of the stellar-dynamical lower limit on the black-hole mass.

The ATCA campaign of Mahida et al. (2026) is deeper on one cluster than the MAVERIC stack is on 24, which is what makes a per-object posterior worth building at all.[2] Three ambiguities in the survey paper are carried rather than resolved: its $\omega$ Cen flux limit appears as 8.8 $\mu$Jy in Table 2 and 8.9 $\mu$Jy in Section V.2.1; its VLA stack mass limit is given as both $< 800$ and $< 730$ $M_\odot$ in a single sentence, and we quote the weaker; and whether $\omega$ Cen enters the 14-cluster ATCA stack is not stated, though the paper's explicit exclusion list omits it. Its comparison of the $\omega$ Cen radio limit against Haggard et al. (2013)

[2]The MAVERIC ATCA catalogue of Tudor et al. (2022) lists a central r.m.s. of 2.8 $\mu$Jy beam$^{-1}$ at 7.25 GHz for $\omega$ Cen; the 1.1 $\mu$Jy of Mahida et al. (2026) is a factor of 2.5 deeper in r.m.s.

**Table 7:** Radio limits on central sources in globular clusters, as published. Mass limits are the sources' own fundamental-plane inversions, quoted at $3\sigma$; they are not $\varepsilon_B$. Flux densities are directly comparable, derived quantities are not.

| Dataset | Depth | Derived limit | Assumed $d$ |
|---|---|---|---|
| MAVERIC ATCA, $\omega$ Cen (Tremou et al., 2018) | $< 8.8\,\mu$Jy ($3\sigma$) | $M < 1000\,M_\odot$ | 4.9 kpc |
| MAVERIC VLA stack, 24 clusters | $0.65\,\mu$Jy beam$^{-1}$ | $M < 800\,M_\odot$ | per cluster |
| MAVERIC ATCA stack, 14 clusters | $1.42\,\mu$Jy beam$^{-1}$ | $M < 970\,M_\odot$ | per cluster |
| This paper's radio input (Mahida et al., 2026) | $1.1\,\mu$Jy beam$^{-1}$ rms | see Fig. 1 | 5.49 kpc |

also cites the X-ray limit as $1.7 \times 10^{30}$ erg s$^{-1}$ where Haggard et al. state $1.6 \times 10^{30}$, which looks like a transcription rounding. None of this affects the numbers of Section 5, which use no MAVERIC input; it affects only what the reader should do with the row above.

# 6 The duty-cycle loophole

The three inputs sample time very differently, and Mahida et al. (2026)'s Table 1 makes the radio sampling explicit: 25 observing blocks across three ATCA projects, 2010 January 22 to 2024 December 27, summing to 177.19 hr on source (CX556, 20 blocks, 147.09 hr; C2877, 3 blocks, 11.79 hr; C2158, 2 blocks, 18.31 hr). Those rows exceed the ∼170 hr of the paper's abstract and the 172 hr of its Section II.3; we report the sum as printed and leave the reconciliation to that paper, noting that prose totals quoted after flagging would run below a raw table sum rather than above it. Chandra contributes four exposures in two epochs twelve years apart, JWST a single epoch.

The consequences differ by timescale. For variability slower than the sampled span, the limits above apply to the time-average $\delta\,\varepsilon_{on}$ of a flow active a fraction $\delta$ of the time at $\varepsilon_{on}$, so $\varepsilon_{on} \lesssim \varepsilon_{95}/\delta$; the radio image is a weighted combination over the full fourteen-year span, so this is the operative statement there. For variability faster than a block, the source must be off during every epoch to escape, which for $N \simeq 30$ independent epochs (25 radio blocks, four Chandra exposures, one JWST visit) gives a miss probability $(1-\delta)^N$ reaching 50 per cent only near $\delta \approx 0.02$. This order-of-magnitude estimate counts every block or exposure as one independent trial regardless of its duration relative to the assumed flare timescale $\tau$; Section 6.1's block-by-block treatment does not carry that assumption and supersedes this estimate in rigor. The excluded region is therefore the band $\varepsilon_{on} > \varepsilon_{95}/\delta$ down to a few per cent duty cycle, and effectively nothing below that: a flow that spends 99 per cent of its time off is constrained only by the luck of the sampling. This is the standard loophole of every quiescent-limit paper, here made explicit because Paper E's tidal-disruption arithmetic predicts such intermittency on $10^7$-yr recurrence with $10^{-4}$ duty, far below the reach of any current sampling; nothing in this paper's nulls bears on that channel, in either direction.

## 6.1 Exclusions set by the block list

The block list supports a stronger statement than a miss probability, because each block is an image in its own right. Scaling the combined image's sensitivity by on-source time, $\sigma_i = 1.1\,\mu\text{Jy}\,\sqrt{177.19\,\text{hr}/t_i}$, the 25 blocks reach 4.5 to 12.0 $\mu$Jy rms with a median of 5.3 $\mu$Jy over a median 7.6 hr on source, so a single block detects a source at $5\sigma$ near 27 $\mu$Jy while the combined image reaches 5.5 $\mu$Jy. That scaling assumes each on-source hour is equally sensitive across the three projects and both observing frequencies, which Mahida et al. (2026) do not state block by block; it is the only scaling their published table supports. The block statistics in this paragraph regenerate from `figs/fF_epochs.json`, the export of that table: 25

**Table 8:** Longest flare recurrence interval excluded at 95 per cent of phases, from the 25-block ATCA timeline (4000 fixed-seed realizations per cell; script `figs/fF_expand_v1.py`, seed 20260820). Entries are hours; a dash means no recurrence interval on the grid is excluded for that amplitude and duration. The implied duty cycle at each threshold, $\tau/T_{\rm rec}$ computed at the unrounded threshold, is in parentheses; recomputing it from the rounded hours in this table gives a value differing by up to 2 per cent. Thresholds are located on a recurrence grid of 45 logarithmic points spanning 6 hr to $2 \times 10^5$ hr, a 27 per cent step, which sets the precision of every entry.

| Flare amplitude | $\tau = 0.5$ hr | 2 hr | 8 hr | 24 hr |
|---|---|---|---|---|
| 10 $\mu$Jy | – | – | 12 (0.655) | 31 (0.763) |
| 30 $\mu$Jy | – | 8 (0.263) | 25 (0.322) | 130 (0.184) |
| 100 $\mu$Jy | 8 (0.066) | 31 (0.064) | 130 (0.061) | 209 (0.115) |
| 300 $\mu$Jy | 15 (0.032) | 81 (0.025) | 130 (0.061) | 209 (0.115) |
| 1 mJy | 81 (0.006) | 81 (0.025) | 130 (0.061) | 209 (0.115) |

blocks spanning 1.48 to 10.70 hr on source, rms 4.48 to 12.04 $\mu$Jy, median 5.31 $\mu$Jy.

A flare train is then a two-parameter object: amplitude $S$ and duration $\tau$, recurring with period $T_{\rm rec}$ at an unknown phase. Laying such a train over the real block timeline (2010 January 22 to 2024 December 27, a 14.9-yr span containing 177.19 hr of on-source time) and asking whether any single block's $\tau$-averaged flux exceeds its own $5\sigma$ point, or the campaign mean exceeds 5.5 $\mu$Jy, gives a detection probability over phase. Table 8 reports, for each $(S, \tau)$, the longest recurrence interval that would still have been caught in at least 95 per cent of phases. Anything recurring faster than that is excluded; anything slower is not.

Low-amplitude trains are excluded only when they are almost always on: a 10 $\mu$Jy source is invisible in any single block and can be caught only through the campaign mean, which requires a duty cycle above 0.7. Bright, brief trains are bounded by the sampling rather than by the sensitivity: at 300 $\mu$Jy and above, for durations of 2 hr and longer, the exclusion saturates near a recurrence interval of 3 to 9 days, because 177 hr of on-source time spread over 14.9 yr covers 0.14 per cent of the span, and a flare that recurs more slowly than about a week simply misses every block most of the time no matter how bright it is. Between those regimes the campaign excludes real parameter space: a 100 $\mu$Jy, 2-hr flare is excluded for recurrence faster than 31 hr, a duty cycle of $6 \times 10^{-2}$.

The scaling's own uncertainty can be priced. Drawing per-block sensitivity log-normally about it, at fixed combined campaign sensitivity, since the 1.1 $\mu$Jy combined rms is measured rather than assumed, moves the table where the single-block channel sets the threshold and leaves it alone where the campaign mean does. At 0.2 dex of scatter, two of the fourteen well-determined cells move by one grid step: the 30 $\mu$Jy, 24-hr threshold falls from 130 to 81 hr and the 100 $\mu$Jy, 2-hr threshold from 31 to 25 hr. Both 10 $\mu$Jy rows hold to the digit at every scatter level tested, up to 0.3 dex, because a 10 $\mu$Jy source is caught only through the campaign mean, whose depth is measured. The floor of the excluded region is likewise unaffected at ±30 per cent per-block rms, and reaches a duty cycle of $1 \times 10^{-2}$ only at 0.3 dex. The caveat this paragraph opens with therefore attaches to the mid-amplitude, long-duration cells rather than to the faint end, which the source paper's combined image already constrains directly. Numbers in `figs/fF_calcF1_duty_band.json`.

The floor of the excluded region is therefore a duty cycle of $6 \times 10^{-3}$, reached at 1 mJy and half an hour, and it is set by sampling coverage rather than by image depth. This tightens the order-of-magnitude estimate above by a factor of a few and it changes the instrument implication: for the flare channel, more epochs at the present depth would buy more than the same time added to a single deep image, which is the opposite of the priority ordering that Section 7 gives for the steady channel. Paper E's predicted tidal-disruption intermittency, at $10^{-4}$ duty on $10^7$-yr recurrence, remains two decades below the floor and six decades beyond the top of the recurrence grid, so the conclusion of the preceding paragraph is

unchanged: this campaign does not constrain that channel.

## 7 Forecasts

Three measurements would harden or move the curves, in descending order of leverage per unit effort.

**A measured core density.** The 19-pulsar timing set (Colom i Bernadich et al., 2026) makes an $\omega$ Cen dispersion-measure gas measurement possible in principle for the first time, by the method introduced on M15's pulsars (Anderson, 1993) and since used to measure 47 Tuc (Freire et al., 2001; Abbate et al., 2018); the fit below shows that the present sight-line scatter prevents it in practice. A second caveat, that DM gradients constrain the column through the pulsar volume rather than the density at the hole, is real and enters as geometry. The forecast gain on $\varepsilon_{95}$ itself is modest (a ±10 per cent density measurement tightens the baseline by ∼22 per cent, since the 0.5-dex prior is already informative). A measurement converts every curve from analogy-conditional to measured, removes the largest single objection to the enterprise, and, if the density comes in low, legitimately weakens the bounds when the conditions are tested.

Table 9 is the requirement that forecast implies, computed from the published timing set. All 19 pulsars have measured dispersion measures; eight have them from full timing solutions with formal uncertainties of $3 \times 10^{-4}$ to $1.2 \times 10^{-3}$ pc cm$^{-3}$, and all eight also carry a fitted first DM derivative, four of them significant at $3\sigma$. Timing precision is not the obstacle.

Geometry sets what a dispersion-measure gradient can reach. Gas filling the cluster core, a uniform sphere of the catalogue core radius 2.37′ of Harris 1996 (2010 edition; Harris, 2010), or 3.78 pc at 5.49 kpc, imprints a central-chord excess of 0.76 pc cm$^{-3}$ at $n_e = 0.1$ cm$^{-3}$, and 13 of the 19 sight lines pass inside it. Gas confined to the influence radius of the hole, the 0.4 pc region the Bondi rate is actually about, imprints 0.08 pc cm$^{-3}$ and no published sight line passes through it: the innermost pulsar, H at 0.56′, projects to 0.89 pc. A DM determination therefore measures the core-filling component and reaches the Bondi-scale density only through an assumed profile, which is the caveat named above, now with a number attached.

The sight-line scatter sets how well it can be measured. The 19 published DMs span 94.3 to 102.6 pc cm$^{-3}$ with an r.m.s. of 2.9 pc cm$^{-3}$, three orders of magnitude above the per-pulsar measurement precision and four times the signal a core-filling $n_e = 0.1$ cm$^{-3}$ would produce. Fitting the chord template plus a free foreground constant to the 19 measurements by least squares returns an amplitude of $-0.33 \pm 0.23$ cm$^{-3}$, consistent with zero and formally negative, with a residual r.m.s. of 2.8 pc cm$^{-3}$. We do not quote the implied upper limit as a constraint on the cluster's gas: a model whose residuals exceed its measurement errors by three orders of magnitude is falsified by its own fit, and the formal error bar of a falsified model is not a credible interval. What the fit does establish is the requirement. At the present per-sight-line scatter, a $3\sigma$ determination of $n_e = 0.1$ cm$^{-3}$ needs the scatter modelled down by a factor of 7, or of order $10^3$ sight lines at the current scatter, which no foreseeable timing programme will deliver. The tractable path is the former: the excess scatter is structure, in the Galactic foreground or in the cluster, and it is measurable in its own right through the DM derivatives that eight of these pulsars already have. Anderson (1993) attributed the M15 dispersion-measure gradient across its seven core pulsars to the foreground interstellar medium on the same evidence of no radial dependence within the core.

**Deeper radio.** The radio-leg limit falls with rms as $\sigma_S^{1/0.60}$ for a single draw, which would give a factor of 46 for a tenfold improvement. The realized marginal gain is much smaller, because the plane's scatter and the other two legs flatten the response: at SKA-Mid-era depth (∼0.1 $\mu$Jy) the RIAF $\varepsilon_{95}$ tightens by 12.8, 11.8 and 5.9 at the three anchors ($3.0 \times 10^{-10} \rightarrow 2.6 \times 10^{-11}$ at 8,200 $M_\odot$), and the exclusion fraction moves from 77 to 94 per cent at the pulsar-timing cap, 85 to 97 per cent at the fast-star anchor and 99 to over 99.9 per cent at $4 \times 10^4$ $M_\odot$.

**Deeper mid-infrared.** Under the RIAF family, deeper MIRI photometry changes nothing: the X-ray

**Table 9:** The 19 timed pulsars as a gas probe. $\theta$ is the angular offset from the cluster centre (Colom i Bernadich et al., 2026), $b$ the projected radius at 5.49 kpc, and $\Delta$DM the excess a uniform core-filling $n_e = 0.1\,\mathrm{cm}^{-3}$ would imprint on that sight line. Pulsars without a quoted DM uncertainty have search-level or partial solutions. Numbers in `figs/fF_expand_dm.json`.

| PSR J1326−4728 | $\theta$ (′) | $b$ (pc) | DM (pc cm$^{-3}$) | $\sigma_{\mathrm{DM}}$ | $\Delta$DM at 0.1 cm$^{-3}$ |
|---|---|---|---|---|---|
| H | 0.56 | 0.89 | 98.1715 | 0.0005 | 0.74 |
| B | 0.76 | 1.21 | 100.2859 | 0.0006 | 0.72 |
| F | 1.00 | 1.60 | 98.29 | – | 0.69 |
| P | 1.00 | 1.60 | 102.17 | – | 0.69 |
| O | 1.50 | 2.40 | 94.305 | – | 0.59 |
| E | 1.58 | 2.52 | 94.3966 | 0.0004 | 0.56 |
| J | 1.80 | 2.87 | 97.28 | – | 0.49 |
| K | 1.89 | 3.02 | 94.7892 | 0.0007 | 0.46 |
| A | 1.93 | 3.08 | 100.3267 | 0.0003 | 0.44 |
| G | 1.96 | 3.13 | 99.7431 | 0.0008 | 0.43 |
| C | 1.98 | 3.16 | 100.6680 | 0.0012 | 0.42 |
| Q | 2.30 | 3.67 | 95.923 | – | 0.18 |
| S | 2.32 | 3.70 | 96.24 | – | 0.15 |
| M | 2.40 | 3.83 | 101.47 | – | 0 |
| D | 2.50 | 3.99 | 96.5467 | 0.0008 | 0 |
| N | 2.66 | 4.25 | 102.1 | – | 0 |
| L | 3.32 | 5.30 | 101.4759 | – | 0 |
| I | 3.53 | 5.64 | 102.555 | – | 0 |
| R | 3.90 | 6.23 | 102.2 | – | 0 |

leg dominates it in every draw, and JWST's role there is the independent-systematics cross-check. Under the jet-dominated family, where the infrared comes closest to mattering, a threefold depth gain moves $\varepsilon_{95}$ by 14 per cent at the fast-star anchor (17 per cent at 6,000 $M_\odot$, 9 per cent at $4 \times 10^4\ M_\odot$). Version 1.5 reported 29, 31 and 21 per cent for the same three numbers (Appendix C). The instrument-priority implication for Paper C is unchanged in order and sharper in degree: radio first, density second, mid-infrared as discrimination between families rather than as depth.

## 8 Scope of the adjudication

**Epistemic status:** Sections 2–7 are instrument-level results and standard accretion arithmetic; nothing in them depends on any hypothesis of this paper set. This section prices the result against the two live readings and is interpretive.

Under the null reading, the center is gas-starved or its flow is suppressed. Under the efficiency law adopted here it is costly everywhere in the contested range: at each of the three anchors it must give up one of its own ingredients, the mass, the fiducial density, or the standard suppression bracket. That is a constraint on conventional astrophysics, publishable and testable on its own terms, and it no longer sharpens the mass tension from one side only, since electromagnetic silence is now hard to buy naturally across the whole range the kinematics argue over. The density is the ingredient this paper's own Section 4 makes it easiest to give up, and doing so is the cheapest way to restore the null reading. Under the engineered reading of Papers A and E, a system extracting accretion power as work rather than radiation presents a deep multi-band null as its expected signature, at any mass; the data cannot separate that reading from the null, and this paper does not claim otherwise. What it contributes to that adjudication is the

currency: a likelihood over $\varepsilon_{\rm B}$ with stated conditioning, which Paper E's framework ingests in place of the single-band, single-convention numbers it previously had to translate. The duty-cycle loophole of Section 6 is charged to both readings equally.

## 9 Conclusion

The deepest radio, infrared, and X-ray observations of $\omega$ Cen's center jointly admit a posterior on the Bondi radiative efficiency of any central accretion flow: $\varepsilon_{95} = 3.0 \times 10^{-10}$ at the fast-star mass anchor for a RIAF-like flow under the stated priors, $1.4 \times 10^{-8}$ with no radio information, and a further factor of 27 tighter at the top of the contested mass range. Scored against a natural outflow-suppressed hot flow at the fiducial density, the data reject 85, 77 and 99 per cent of that flow's parameter space at the fast-star anchor, the pulsar-timing cap and $4 \times 10^4\,M_\odot$, and above 0.57, 0.47 and 0.93 across every density prior and convention of Table 6. The curve's normalization is set by the unmeasured central density, and the cluster's own stellar winds now estimate that density well below the analogy value: a recount of the giants inside the influence radius gives eight of them and a no-removal steady state near $0.03\,\mathrm{cm}^{-3}$ at the assumed outflow speed, with 0.03–0.08 $\mathrm{cm}^{-3}$ across the estimates of Section 4, a factor of three to eight under the 47 Tucanae value the prior is centred on. That recount points the density evidence towards the weaker end of the exclusion range even as the efficiency correction points the verdict towards the stronger. A pulsar dispersion-measure gradient would settle the density, but at the present sight-line scatter that fit is foreground-limited: the 19-pulsar set returns $-0.33 \pm 0.23\,\mathrm{cm}^{-3}$ against residuals about $10^3$ times the measurement errors, and a $3\sigma$ determination needs the scatter modelled down by a factor of 7, or of order $10^3$ sight lines (Section 7). Until that scatter is modelled, the density stays an analogy-conditional input. In the meantime, Figure 1 is the quantitative meaning of $\omega$ Cen's silence, and the engineered-silence hypothesis of this paper set lives inside it on equal terms with a gas-starvation null that must now give up one of its own ingredients at every anchor rather than only at the highest.

## Data availability

The anchors, limits and exclusion fractions printed at 5.49 kpc are produced by `figs/plc_f/run_plc_f.py` and stored in `figs/plc_f/out/distance_549.json`; the 5.43 kpc run is retained as `figs/fF_v4_results.json` for comparison. The posterior script (`figs/fF_posterior_v4.py`, fixed seed), its full output (`figs/fF_v4_results.json`), its seed study (`figs/fF_v4_seedspread.json`) and realized-scaling export (`figs/fF_v4_scalings.json`), the standalone recomputes behind the corrections of this revision (`figs/fF_r9_checks.py` and its output), the superseded v0.3 script and output retained unchanged as the previous computation of record, the exclusion-fraction tables, the measured-input provenance file that supplies every observational number used above with its source location and verbatim quotation (`figs/fF_measured_inputs.json`) together with the derived epoch and survey-context exports (`figs/fF_epochs.json`, `figs/fF_maveric_context.json`), the superseded v0.2 script and output retained for comparison, the appendix cross-check script (`figs/fF_joint_bound.py`) and its output (`figs/fF_results.json`), the expansion-analysis scripts behind Figures 3, 4 and 5 and Tables 8 and 9 (`figs/fF_expand_v1.py` for the duty-cycle and dispersion-measure analyses, which read no likelihood and are unchanged, and `figs/fF_expand_v2.py` for the three products that do, with outputs `fF_expand_nuisance_v2.json`, `fF_expand_sed_v2.json`, `fF_expand_exclcurve_v2.json`, `fF_expand_duty.json` and `fF_expand_dm.json`), and the figure coordinate tables are distributed with the paper source at omegacentauri.me. The pulsar dispersion measures of Table 9 are read from the timing-set extraction `h/data/pulsars.json`, which records each value's source location in Colom i

Bernadich et al. (2026). The provenance file also records two internal inconsistencies in the sources that a reader checking our inputs will meet: Mahida et al. (2026)'s conservative efficiency limit is stated as $4 \times 10^{-3}$ in their abstract and Section IV.1 and as $4 \times 10^{-6}$ in their Section V, as of arXiv v1, with the Section IV.1 derivation itself supporting the $4 \times 10^{-3}$ figure, the adiabatic limit once fundamental-plane scatter is included ($8 \times 10^{-5}$ adiabatic and $9 \times 10^{-7}$ isothermal without it, $4 \times 10^{-3}$ and $5 \times 10^{-5}$ with); and their Table 1 observing hours sum to 177.19 hr against prose totals of 170 and 172 hr (Section 6). Neither enters any number in this paper. The analysis would improve with, and the machinery directly ingests, the measured central-pixel flux density and its uncertainty from the ATCA image, the per-epoch Chandra source counts and background at the center list, and the per-filter MIRI/NIRCam limiting fluxes.

# A Conventions, unit conversions, and cross-paper reconciliation

**Bondi rate.** Eq. 1 uses the $\gamma = 5/3$ eigenvalue $\lambda = 0.25$ (Bondi, 1952). A constant copied from the 1952 numerics belongs to the same error species as the $\xi$ correction recorded in Paper H, so we state how it verifies. Bondi's general polytropic eigenvalue, valid for $1 < \gamma < 5/3$, is

$$\lambda(\gamma) = \frac{1}{4}\left[\frac{2}{5-3\gamma}\right]^{(5-3\gamma)/2(\gamma-1)}, \tag{10}$$

which evaluates to 0.500 at $\gamma = 1.5$, 0.625 at $\gamma = 1.4$ and 1.120 as $\gamma \to 1$, reproducing the tabulated values, while as $\gamma \to 5/3$ the bracketed factor tends to unity and leaves the prefactor 1/4 (see also Frank et al., 2002). Up to v1.5 this equation was printed with a spurious second factor (Appendix C); it is removed here, the values above are recomputed in `figs/fF_r9_checks.py`, and no number elsewhere in this paper used it: $\dot{M}_{\rm B}$ was always evaluated at $\lambda = 0.25$. At $\gamma = 5/3$ itself the regular sonic point of the general solution merges with the origin, so the value is read as this limit rather than from a critical-point evaluation. The rate uses a gas-rest-frame denominator $c_s^3$, with $c_s$ drawn between the $\mu = 0.6$ isothermal value (11.7 km s$^{-1}$) and the pure-hydrogen adiabatic value (16.6 km s$^{-1}$) at $10^4$ K. Paper E's fuel budget (its Eq. 2) uses $\lambda = 1$ with a $(\sigma^2 + c_s^2)^{3/2}$ denominator as a turbulent-medium convention; the net ratio between the two conventions is $0.25\,(\sigma^2 + c_s^2)^{3/2}/c_s^3$, which is 1.43 at the median sound speed $c_s = 14.15$ km s$^{-1}$ and reaches 1.9 only at $c_s \simeq 12.3$ km s$^{-1}$, the 13th percentile of the prior. The 1.9 figure and $\dot{M}_{\rm B} = 5.3 \times 10^{18}$ g s$^{-1}$ at $2 \times 10^4\,M_\odot$ quoted as median values in v1.5 both belong to that 13th-percentile sound speed. At the median the rate is $\dot{M}_{\rm B} = 3.5 \times 10^{18}$ g s$^{-1}$ against Paper E's $2.5 \times 10^{18}$ evaluated from its Eq. 2 at the same median sound speed (Paper E prints its rate at its own sound speed of 11.5 km s$^{-1}$, where the same equation gives $2.9 \times 10^{18}$; at the sound speed used here it gives $2.48 \times 10^{18}$, a ratio of 1.41), so this paper's $\dot{M}_{\rm B}$ is 1.4× Paper E's at equal $(M, n_e)$ under median nuisances and its $\varepsilon_{\rm B}$ limits are correspondingly 1.4× tighter than under E's convention. Both statements are correct within their stated conventions; the correction is to the exchange rate, not to either paper's own numbers.

**Electron mean molecular weight.** $\mu_e = 2/(1 + X) = 1.17$ at $X = 0.71$.

**Radio.** $L_R \equiv \nu L_\nu$ at 5 GHz under a flat spectrum: $L_R = 4\pi d^2\,(5\,{\rm GHz})\,S_\nu$, with $S_\nu$ the 7.25 GHz flux density ($L_\nu$ flat). At $d = 5.49$ kpc, 1.1 $\mu$Jy corresponds to $L_R = 2.0 \times 10^{26}$ erg s$^{-1}$ ($5.9 \times 10^{26}$ at $3\sigma$).

**X-ray bands.** The fundamental plane is calibrated on 2–10 keV; the Chandra limit is 0.5–7 keV; $F(2\text{–}10)/F(0.5\text{–}7) = \ln(10/2)/\ln(7/0.5) = 0.610$ for $\Gamma = 2$, which is the value used throughout. Haggard et al. (2013) derive their own limit under $\Gamma = 2.3$, and at that index the ratio of power-law integrals is 0.462, lower by 24 per cent. The radio leg enters as $L_R \propto (k f_X L_{\rm bol})^{0.60}$, so the change moves the predicted radio luminosity by 0.85 and loosens the RIAF anchors by 15 to 20 per cent (Table 10), the largest single convention effect in this appendix. It is carried as a ladder row rather than adopted, because

the plane's own calibration sample is not restricted to $\Gamma = 2.3$ sources and changing $k$ alone would mix the two calibrations.

**X-ray reference distance.** Haggard et al. (2013) adopt 5.2 kpc (their abstract and Table 1, from the Harris catalogue, which prints 5.2 kpc in Harris 1996 (2010 edition; Harris, 2010)). The mismatch against the 5.49 kpc prior shifts the X-ray leg by nothing: the X-ray likelihood is evaluated on flux, which is what Chandra measured and is distance-independent, so the drawn distance enters the predicted flux and never the limit. The reference distance serves as a consistency check on the source's own numbers: $4\pi D^2 f_{X,\mathrm{lim}} = 1.62 \times 10^{30}\,\mathrm{erg\,s^{-1}}$ at 5.2 kpc against the $1.6 \times 10^{30}$ Haggard et al. (2013) state, which holds at their distance. The script asserts it at run time. The infrared limits, by contrast, are published as luminosities and do carry their source distance, so they are referenced at Chen et al.'s 5.49 kpc and rescaled draw by draw.

**Run-time self-checks.** Consistency assertions accompany the shipped code and print their result rather than assuming it, the discipline of Paper H's self-audit. The expansion script (`figs/fF_expand_v2.py`) reproduces the v0.4 RIAF anchors of `fF_v4_results.json`, drawn at 5.43 kpc, and asserts the match before proceeding; the anchors of Table 4 are the 5.49 kpc values produced by `figs/plc_f/run_plc_f.py` and stored in `figs/plc_f/out/distance_549.json`. The Haggard luminosity-anchor check above asserts $4\pi D^2 f_{X,\mathrm{lim}}$ against Haggard et al.'s stated $1.6 \times 10^{30}\,\mathrm{erg\,s^{-1}}$ at their own distance. And the v0.4 posterior asserts, before computing anything of its own, that with its four corrections disabled it reproduces the v0.3 output file to floating-point equality, so the refactor cannot have moved a number by itself (Appendix C). All three halt the build on failure, so a compiled paper carries their pass by construction.

**Confidence conventions.** Each leg uses the confidence its source states, mapped through the normal quantile. The X-ray limit is one-sided 95 per cent ($z = 1.645$); the infrared limits are taken at the 95 per cent completeness row on the same convention. Neither choice is load-bearing:

**Table 10:** Convention ladders for the two limit legs, plus the X-ray band-conversion index. RIAF-family $\varepsilon_{95}$ at the three mass anchors; the primary rows are those used throughout the paper.

| Leg | Convention | 6,000 $M_\odot$ | 8,200 $M_\odot$ | $4 \times 10^4$ $M_\odot$ |
|---|---|---|---|---|
| X-ray | one-sided 95 % ($z$=1.645, primary) | $6.25 \times 10^{-10}$ | $3.01 \times 10^{-10}$ | $1.10 \times 10^{-11}$ |
| X-ray | two-sided 95 % ($z$=1.960) | $6.15 \times 10^{-10}$ | $2.97 \times 10^{-10}$ | $1.09 \times 10^{-11}$ |
| X-ray | lim/3 ($3\sigma$ reading) | $5.87 \times 10^{-10}$ | $2.85 \times 10^{-10}$ | $1.07 \times 10^{-11}$ |
| IR | 99.7 % completeness row | $6.27 \times 10^{-10}$ | $3.02 \times 10^{-10}$ | $1.10 \times 10^{-11}$ |
| IR | 95 % completeness row (primary) | $6.25 \times 10^{-10}$ | $3.01 \times 10^{-10}$ | $1.10 \times 10^{-11}$ |
| IR | 68 % completeness row | $6.19 \times 10^{-10}$ | $2.99 \times 10^{-10}$ | $1.09 \times 10^{-11}$ |
| IR | tightest single filter, 95 % | $6.26 \times 10^{-10}$ | $3.02 \times 10^{-10}$ | $1.10 \times 10^{-11}$ |
| X-ray band | $\Gamma = 2$ ($k = 0.610$, primary) | $6.25 \times 10^{-10}$ | $3.01 \times 10^{-10}$ | $1.10 \times 10^{-11}$ |
| X-ray band | $\Gamma = 2.3$ ($k = 0.462$, the source's) | $7.52 \times 10^{-10}$ | $3.61 \times 10^{-10}$ | $1.26 \times 10^{-11}$ |

The X-ray ladder spans 3 to 7 per cent on these anchors and the infrared ladder 1 per cent, the latter having narrowed with the band correction of §2 because the infrared terms now sit further from binding. On the jet family, where the infrared leg comes closest to mattering, the tightest-single-filter reading gives $1.43 \times 10^{-9}$, $6.76 \times 10^{-10}$ and $2.13 \times 10^{-11}$ against the combination's $1.40 \times 10^{-9}$, $6.65 \times 10^{-10}$ and $2.11 \times 10^{-11}$, a 2 per cent effect, because F770W dominates the combination in any case. Referencing the infrared limits at 5.43 kpc, the earlier distance centre, instead of Chen et al.'s 5.49 kpc moves the jet anchors by at most 0.3 per cent.

**Reconstruction of the limit legs.** A published bound states $P(\mathrm{observed} < f_{\mathrm{lim}})$, which is a survival statement, while Section 3.3 evaluates a Gaussian centred at zero with the same $\sigma$. Substituting the

survival form on the X-ray and infrared legs, the two published as limits with no central value, moves the radio-free and thin-disk anchors by 1 per cent and leaves every exclusion fraction unchanged to three decimals; those two curves carry only those legs, so that is the size of the choice wherever it applies alone. With the infrared $\sigma_j$ a factor of four to six larger after the band-integration correction of §2 (v1.5 put this figure at 29 to 30 per cent), the infrared terms sit far enough from their limits that the two likelihood shapes agree, which is the regime in which they are supposed to. The radio leg is excluded from the substitution because Mahida et al. (2026) report a measured central flux density rather than a bound, and a survival form there would discard the measurement; doing so anyway loosens the RIAF anchors by a factor 3.2 to 3.9 and the jet anchors by 2.8 to 3.0, which is the value of the measurement rather than a property of the reconstruction. An injection study of 200 realizations per cell puts the frequentist coverage of the 95 per cent credible upper limit at 0.92 at 8,200 $M_\odot$ and 0.96 at $4 \times 10^4\ M_\odot$ when the true $\varepsilon_B$ sits at the published limit, against a binomial uncertainty of 0.02. A credible interval carries no coverage guarantee, and the check is run at one point in $(\varepsilon_B, M)$ rather than over the parameter space, so it is a calibration spot-check of the reconstruction convention rather than a coverage statement about the method. Numbers in `figs/fF_calcF1_v2_partc.json`, recomputed on the v0.4 likelihood; the v0.3 values are retained in `figs/fF_calcF1_hybrid.json` and `figs/fF_calcF1_coverage.json`.

**Cross-paper exchange rates.** At the common reference point ($M = 2 \times 10^4\ M_\odot$, $n_e = 0.23\,\mathrm{cm}^{-3}$, median nuisances): this paper's $\varepsilon_B$ relates to the horizon-side efficiency as $\eta = \varepsilon_B / f_B$ with $f_B \in [10^{-4.4}, 10^{-2.6}]$, the bracket of Section 3.1 under $r_g = GM/c^2$; Mahida et al. (2026)'s bounded quantity is an accretion fraction under their assumed radiative model and is neither $\varepsilon_B$ nor $\eta$; Haggard et al. (2013)'s efficiency is an $\varepsilon_B$-like quantity evaluated at their own inputs, and those inputs are now stated here rather than left implicit: $M = 1.8 \times 10^4\ M_\odot$, $n_e = 0.038\,\mathrm{cm}^{-3}$ from a Pfahl–Rappaport wind budget, and $T = 10^4$ K, giving the $\eta = 4 \times 10^{-9}$ to $1.3 \times 10^{-8}$ they compute for a simple Bondi rate. Because $\varepsilon_B \propto 1/(n_e \lambda)$ at fixed luminosity, the exchange factor into this paper's convention is $(0.038\,\lambda_H)/(0.23 \times 0.25)$: at $\lambda_H = 0.25$ it is 0.17, and it reaches 0.66 only if their rate is read with $\lambda_H = 1$. The factor is 0.17–0.66 (v1.5 printed 0.5–1.5, which no reading of the two densities supports) depending on the $\lambda$ convention attributed to that paper, and the residual ambiguity is theirs to resolve. The density difference is the substance of it: their $0.038\,\mathrm{cm}^{-3}$ is a factor 6.1 below this paper's prior median, and §4 now reaches a comparable figure from an independent route. The same caution applies to Mahida et al.'s accretion fraction, whose model normalization is not stated in convertible form. Until both are re-evaluated in one convention, the qualitative ordering (radio strongest, X-ray next, IR family-dependent) is what transfers.

## A.1 Shared fiducial parameters across the paper set

Table 11 is a cross-reference index, not a mandate: it records the value each paper in the set actually uses for seven recurring cluster parameters, and where a paper's choice differs from the majority it names the deviation rather than silently harmonizing it. No paper's own derivation is changed by this table; where two papers differ, both retain their own number, and the table exists so a reader moving between papers can see the exchange rate rather than assume equality (R9-XPAPER-1 X-R9-08/09/10/11, folding R9-SSOT-1 and the R8 leftover X-R8-13 on core-radius conventions).

## A.2 $\sigma$ fiducial disclosure

This paper's likelihood does not use $\sigma$ directly, but the flyby and dynamical calculations elsewhere in the set do, at the $21\,\mathrm{km\,s}^{-1}$ paper-set fiducial rather than the $18.2\,\mathrm{km\,s}^{-1}$ the public MCP calculators default to; the two are not harmonized, and the $(21/18.2)^2 \approx 1.3$ factor in $r_{\mathrm{infl}}$ is the size of the resulting offset (X-R9-12, the same disclosure Paper E already carries at its own flyby-envelope figures).

| Quantity | Value used, by paper (deviation noted where it exists) |
|---|---|
| Distance $d$ | A/C/E/G/H: 5.49 kpc (Häberle et al., 2025) (H uses 5.494 kpc to three decimals; G's angular catalogue does not carry a distance). D follows the same 5.49 kpc set fiducial, and so does F as its stated fiducial; B states no cluster distance. F's distance prior is centred on it (§3.2); versions up to 2.5 carried numbers at 5.43 kpc (Baumgardt and Vasiliev, 2021). |
| Velocity dispersion $\sigma$ | Paper-set fiducial $\sigma = 21\,\mathrm{km\,s^{-1}}$ (e.g. Paper E's flyby Monte Carlo, its Appendix A). The public MCP calculators default to $18.2\,\mathrm{km\,s^{-1}}$ instead; $(21/18.2)^2 \approx 1.33$ in $r_{\mathrm{infl}} \propto \sigma^{-2}$, so a reader moving between a paper figure and a calculator output should expect a ~33 per cent offset in influence radius from this choice alone (§A.2 below; the same disclosure Paper E already carries). |
| Core radius $r_c$ | Not one quantity: three different conventions appear under the same symbol and are not interchangeable. A/F: 2.37′, the King-model core radius in Harris 1996 (2010 edition; Harris, 2010). G: 155″ (= 2.58′), its own catalogue core radius (Anderson & van der Marel 2010), used as the reference scale for source offsets and cluster-membership weighting. E: 3.6 pc physical (≈ 2.25′ at 5.49 kpc), the adopted flat-core radius of its enclosed-mass model. The three differ by up to ~15 per cent in angular terms and are stated here with their conventions attached rather than forced to one number. |
| Cluster mass $M_{\mathrm{cl}}$ | A: $\approx 4 \times 10^6\,M_\odot$ (Harris, 1996; Baumgardt and Vasiliev, 2021), its adopted catalogue total (its own parameters table). E: $3.55 \times 10^6\,M_\odot$, the Plummer-profile total mass used in its enclosed-mass model. H's own visible-mass fit gives $M_{\mathrm{vis}} = 3.718 \times 10^6\,M_\odot$, 4.7 per cent above E's figure and not with A's catalogue total, which is not itself a dynamical-model input in A. |
| Central density $\rho_c$ | A: catalogue value $\sim 3 \times 10^3\,M_\odot\,\mathrm{pc^{-3}}$ (Baumgardt and Vasiliev, 2021), alongside its own adopted King-model value $4.0 \times 10^3\,M_\odot\,\mathrm{pc^{-3}}$ (33 per cent above the catalogue figure; both are disclosed side by side in its own parameters table). C/E: adopted flat-core value $3 \times 10^3\,M_\odot\,\mathrm{pc^{-3}}$, matching A's catalogue row rather than its King-model row. |
| Mean/effective stellar mass $\langle m \rangle$ | A: $0.43\,M_\odot$ per star, the conversion used for its stellar-count enclosed-mass table. E: $0.425\,M_\odot$, the mean of its two-component baseline mass function with no remnants (its flyby-Monte-Carlo appendix); the two differ by < 2 per cent and are not reconciled further here. |
| Electron density $n_e$ | F's own wind-recount work (§4): prior median $0.23\,\mathrm{cm^{-3}}$ (log-normal, 0.5 dex, 47 Tuc analogy; Freire et al. 2001; Abbate et al. 2018), independently estimated by F's stellar-wind budget at $\sim 0.029\,\mathrm{cm^{-3}}$ at the assumed $v_{\mathrm{out}} \sim 50\,\mathrm{km\,s^{-1}}$, within a 0.03–0.08 $\mathrm{cm^{-3}}$ bracket across the estimates of §4 (an estimate, not a ceiling); no other paper in the set adopts an independent $n_e$ value. |

**Table 11:** Shared fiducial parameters, by paper, as each paper's own text currently states them. Not a harmonization: differences are cross-referenced, not resolved, except where a paper's own revision (outside this WU's scope) already brought it into line.

## B Threshold-inversion cross-check

An earlier construction of this bound (script `figs/fF_joint_bound.py`, retained unchanged with its own seed and output) inverted the per-band thresholds as that draft labelled them, $3\sigma$ in every band, through the same nuisance draws and reported percentiles of the per-draw minimum. It is transparent and reproducible, and its 95th-percentile "conservative" values ($1.5 \times 10^{-8}$ at 8,200 $M_\odot$ with the radio leg; $3.0 \times 10^{-7}$ without) sit one to two decades above the credible limits of Table 4, for two reasons with opposite signs: the threshold construction discards the information in the measured values (weakening), and its percentile is a nuisance-prior quantile of a limit rather than a posterior statement about $\varepsilon_{\rm B}$ (not comparable in coverage). The disagreement is largest where the fundamental-plane tail dominates, as expected. Its X-ray threshold also predates the confidence relabelling of Section 2, which is a further reason the two constructions are not expected to agree numerically. The posterior supersedes it; it remains in the distribution as a check that no single modeling choice in Section 3.3 drives the headline numbers by itself.

## C Revision record

This appendix lists what changed in each version, so that a number quoted from an earlier version can be traced to its replacement. Entries give the superseded value first.

- **v2.11 (2026-11-02, style pass).** Style pass under the series' prose rules; no number, equation, citation, table entry, figure, claim or hedge changes. Four headings in question or "What/Where" form are restated: §5.2 is now "The binding band in each family", and the nuisance subsection of §5.3, the survey-population subsection of §5 and the block-list subsection of §6 are renamed to match. Most numeric-announce openers are rewritten, leaving three, and anthropomorphic phrasing ("owns", "travels with", "says nothing", "pays a price") is rewritten as plain statements. The run-in sentence labels and the one defined term keep their emphasis.

- **v2.10 (2026-11-02, two held items closed).** Two items held out of the v2.9 audit, applied on Tim's ruling of 2026-11-01 after they were read against the pulsar-timing source (farm D574, applied in D584); no other number moves. The dispersion-measure paragraph of §7 said that five of the eight pulsars with full timing solutions also have a first DM derivative; all eight do, since a solution that fits the derivative reports it whether or not the fit is significant, so the clause now reads all eight and carries the significance it withholds, four of them at $3\sigma$, counted from the v2 posting of the timing paper, whose appendix tables revise the derivative values and their errors for all eight pulsars (I4, first site). The back-reference later in the same paragraph read five, and now reads eight (I4, second site). Table 5 gains a last row pricing Paper E's Bondi convention; its three cells, $8.2 \times 10^{-10}$, $3.9 \times 10^{-10}$ and $1.3 \times 10^{-11}$, are the same anchors under that denominator, 1.32, 1.30 and 1.22 times the baseline, and are taken from the already-computed variant rather than recomputed here (I3). The row keeps $\mu_e = 1.17$, so it prices the $\lambda$ and denominator difference alone. No limit, exclusion fraction, figure coordinate or other table entry changed. Four loose ends of the same pass are closed (farm D597, checked in D598). The parenthetical of §3.1 that explained the $3 \times 10^{18}$ Paper E once printed is replaced: Paper E prints its rate at its own sound speed of 11.5 km s$^{-1}$, where Eq. 2 gives $2.9 \times 10^{18}$, and at the sound speed used here it gives $2.48 \times 10^{18}$, so the quoted $2.5 \times 10^{18}$ and the ratio of 1.4 stand. Table 9 and the precision range of §7 were still on the v1 timing cells of the same paper; the eight DM values and their errors now follow the v2 tables (H 98.1715, B 100.2859, E 94.3966, K 94.7892, G 99.7431, C 100.6680, D 96.5467, A unchanged), the formal uncertainties

read $3 \times 10^{-4}$ to $1.2 \times 10^{-3}$ (was $5 \times 10^{-5}$ to $1.3 \times 10^{-3}$), and the span, r.m.s., fitted amplitude and sight-line count of that paragraph are unchanged on recomputation. `figs/fF_expand_dm.json` is re-exported from the v2 cells; its note on pulsar Q now states that Table 1 of the timing paper prints 95.955 and Table A.3 prints 95.923, 0.032 pc cm$^{-3}$ apart and not a rounding, and that this paper uses the A.3 value.

- **v2.9 (2026-11-01, citation and constant fixes).** Three corrections from the citation-and-constant audit of this paper and its cross-check (farm D531 and D534). None moves a number in the abstract, the conclusions or a table. The fast-star aperture of the introduction reads the central three arcseconds for Häberle et al. (2024), in place of the single arcsecond printed earlier, with the 8,200 $M_\odot$ limit unchanged (F1). The mass range over which Chen et al. (2026) state their radio comparison is 6,000 $M_\odot$, so the $4 \times 10^4$ $M_\odot$ anchor lies a factor of about seven above the range they state, against the four printed earlier (F2). The Bondi-convention variant of Section 3.1 lowers $\dot{M}_{\rm B}$ by 1.4 at the median sound speed and reaches 1.9 only at the 13th percentile of the prior, which is what Appendix A and the v2.0 entry already record (I1). No limit, exclusion fraction, figure coordinate or table entry changed.

- **v2.8 (2026-10-27, figure-text and record reconciliation; first pass 2026-10-25).** Four figure and table defects that a two-pass audit of figures and tables against prose found (farm D312 and its cross-check) are repaired at the paper's adopted distance. Figure 4 and its record `figs/fF_expand_sed_v2.json` were rebuilt on the 5.49 kpc draw prior (`research/fix-f-figs-549/scripts/re` which imports the shipped modules and promotes the distance median to an argument, the method of the v2.4 rerun): the RIAF $\varepsilon_{95}$ at 8,200 $M_\odot$ moves from $2.94 \times 10^{-10}$ to $3.01 \times 10^{-10}$, the value Table 4 and Table 10 already printed, and the ATCA $5\sigma$ conversion moves from $1.41 \times 10^{27}$ to $1.44 \times 10^{27}$ erg s$^{-1}$ with the distance. Three ratios quoted in §5.2 follow: the jet radio ratio at $4 \times 10^4$ $M_\odot$ reads 3.5 (was 3.6) and the thin-disk X-ray fractions read 0.44 and 0.59 (were 0.43 and 0.58). The four coordinate arrays of Figure 1 were regenerated on the same prior and move by at most 3.2 per cent, with the natural-flow band unchanged because $\varepsilon_{\rm nat}$ carries no distance. Figure 2 was rebuilt with the $5.49 \pm 0.05$ kpc prior of Table 2, so its distance panel is centred on the paper's value; its effective sample size reads $1.06 \times 10^5$ (was $1.05 \times 10^5$) and its provenance now names `fF_posterior_v4.py`, the module it imports, in place of the superseded v3. Table 11 is cited where Appendix A.1 introduces it. Widening the $n_e$ prior costs the RIAF exclusion fraction 0.10 at the fast-star anchor (was 0.09), the value Table 6 gives. No limit, exclusion fraction, or table entry other than those named changed.

  A second pass over the record itself, and over three numbers this paper quotes from its companions (farm D318 and D319), corrects what the earlier entries say about their own commits. The v2.0 entry's linear-$\eta$ parenthetical, printed at 0.48, reads 0.47, the value the row carries now. The v2.4 entry records the relabelling of the sensitivity row from the lower edge of the Xie and Yuan (2012) lowest branch to its upper edge, and its anchor count moves from seven of the eighteen printed entries to eight. The v2.5 entry's two corrected bibliography entries become one, the Spitzer author field, plus a comment block removed from a bib file this paper does not load. The v2.6 entry records the superseded-count factor moving from 119 to 123 and the Haggard comparison from 30 to 32 per cent. The v2.7 entry attributes the crossing statement to the limits figure rather than to the exclusion curve. The distance appendix reads versions up to 2.5 at 5.43 kpc, which is where the v2.6 entry says the three groups of numbers stood. Paper E's suppression, quoted at 500, reads 290, which is what E derives from its own $r_{\rm B}$ at $2 \times 10^4$ $M_\odot$. Paper H's visible-mass fit is quoted at the superseded FIT-1 value of $3.54 \times 10^6$ $M_\odot$ and a sub-per-cent agreement with Paper E; it reads

$3.718 \times 10^6\ M_\odot$, 4.7 per cent above E's $3.55 \times 10^6$. The shared table no longer attributes a 5.49 kpc distance to Paper B, which states none, and Paper A's catalogue-to-King density offset, printed at 34 per cent, reads 33, which is 4.0 against 3.0 at printed precision. No limit, exclusion fraction, figure coordinate or table entry changed in this pass.

A third pass (farm D359) brings the header comment and this entry into agreement and clears the one construction that stopped the file from building. The comment block's parenthetical reads as this entry's heading does. The citation attached to the Bondi eigenvalue in Eq. 1 is set in text mode, which is what TeX Live 2023 accepts inside a display; no number, table entry or figure coordinate changed. The decade-down prior shift of §5.3 prints 0.27 because the unrounded exclusion fractions give 0.273575, against the 0.28 that subtracting the two printed cells of Table 6 would give.

- **v2.7 (2026-10-24, R10 fact pass).** Sound-speed rows added to Tables 5 and 6 (bracket ends and $10^5$ K; ratios of $\varepsilon_{95}$ to baseline 0.64, 0.65, 0.72 at the low end, 1.45, 1.42, 1.31 at the high end, and 9.7, 9.2, 5.9 at $10^5$ K, computed by the new `figs/fF_cs_sensitivity.py`), with the justification of the $c_s$ centre in Table 2 and the sound-speed sentence of §5.3.1 reworded to separate scatter from centre. Prose that disagreed with the tables was corrected: the 90 per cent RIAF threshold ($1.1 \times 10^4\ M_\odot$) lies above the point-mass cap, not below it; the $\eta$-flat row gives 0.87 and 0.92 for the radio-free entries at the two lower anchors; the infrared sits five to six times below the X-ray in the RIAF and thin-disk families and 1.2 times above it in the jet family; the limits figure's caption states where the band median and 5th-percentile edge cross the RIAF limit; the grid-floor remark reads factors of 9 and 11 at the two lower anchors (was a factor of 9); the radio-free range is 0.04 to 0.39 at the fast-star anchor (was 0.02) and 0.32 to 0.82 at the high anchor (was 0.80); the intermittency remark reads six decades beyond the top of the recurrence grid (was 12 decades outside the recurrence range); and Paper E's accretion rate is evaluated from its Eq. 2 at the same median sound speed ($2.5 \times 10^{18}$, was $3 \times 10^{18}$). The run-time self-check paragraph and the data-availability paragraph now state that `fF_posterior_v4.py` and `fF_v4_results.json` are the 5.43 kpc v0.4 computation and that the 5.49 kpc anchors come from `figs/plc_f/run_plc_f.py`. The Harris catalogue is cited as Harris 1996 (2010 edition). No headline number changed.

- **v2.6.1 (2026-10-23).** The error on the fitted dispersion-measure amplitude in the conclusion is $0.23\ \mathrm{cm}^{-3}$ at 5.49 kpc, matching Section 7; v2.6 had left the earlier 0.24 there.

- **v2.6 (2026-10-23).** The three groups of numbers left at 5.43 kpc in v2.4 and v2.5 were rerun at 5.49 kpc (farm D266, same code, distance median the only change) and none remains at the earlier centre: in the dispersion-measure section the core radius is 3.78 pc at 5.49 kpc, the central-chord excess 0.76, the projected radius of pulsar H 0.89 pc and the error on the fitted amplitude $0.23\ \mathrm{cm}^{-3}$, with Table 9 updated throughout; in Table 6 the linear-$\eta$ row reads 0.47 for the RIAF family at 8,200 $M_\odot$ and 0.26 without radio at $4 \times 10^4\ M_\odot$, and the forecast fractions of §7 do not change at printed precision. The Plummer scale length of the wind recount is the catalogue core radius of 2.37 arcmin at the distance, so it moves from 3.74 to 3.78 pc at 5.49 kpc, and the recount follows (`figs/fF_calc7_windfloor.py`, rerun): 8.1 standing giants in place of 8.4 (5.2 to 11.0), an injection of $8.1 \times 10^{-8}\ M_\odot\ \mathrm{yr}^{-1}$ and a steady-state density of $0.029\ \mathrm{cm}^{-3}$ in place of 0.030 (0.018 to 0.039 across the variations). The same rerun moved the superseded-count factor from 119 to 123 and the Haggard comparison from 30 to 32 per cent. The statement that the disk-family limit sits four to five decades below the disk efficiency is corrected to six to nine: $\log_{10}(0.06/2.4 \times 10^{-8}) = 6.4$ to $\log_{10}(0.4/7.4 \times 10^{-10}) = 8.7$ from Table 4.

- **v2.5 (2026-10-23).** Catalogue attribution corrected: the core radius of 2.37 arcmin is the King-model

value in the 2010 revision of the Harris catalogue (cited as Harris 2010 in Appendix A and in the dispersion-measure geometry), and the 5.2 kpc that Haggard et al. (2013) adopt is stated as the distance that revision prints, without a claim about which edition they used. One bibliography entry was corrected, the Spitzer author field in `references.bib`; a comment block was also removed from a bib file this paper does not load. No number changed.

- **v2.4 (2026-10-22).** The distance prior was recentred on the set fiducial of 5.49 kpc (§3.2), with the width of 0.05 kpc kept, and the posterior was rerun on the same stream. Every $\varepsilon_{95}$ rose by 1.2 to 3.1 per cent and every exclusion fraction fell by up to 0.0033. Eight of the eighteen printed entries of Table 4, twenty-one of the twenty-four of Table 5, eleven printed fractions of Table 6, the abstract's 78 per cent (now 77) and its 58 and 33 per cent (now 57 and 32) changed at the printed precision. Table 6 gained rows at prior medians of 0.08 and 0.03 $\mathrm{cm}^{-3}$, and Table 10 was rerun. The statement that the lowest rate computed by Xie and Yuan (2012) is $1.6 \times 10^{-5}$ to $9.4 \times 10^{-5}$ was corrected: those are the upper edges of the lowest branch of their Table 1, which prints no lower edge, so the extrapolation depth is measured below that edge. The sensitivity row that holds $\eta$ at that edge was relabelled with the same commit: it reads the fit's own upper edge, the top of its lowest branch, where it previously read the lower edge, and the prose introducing the row follows it.

- **v2.3 (2026-10-20).** Retitled as a radio and X-ray bound with an infrared consistency check, since the infrared leg never binds (§5.2). The abstract was shortened and the claim that a dispersion-measure determination is feasible with the current 19-pulsar set was removed to match §7. The wind-supply budget is described as an estimate linear in the assumed outflow speed, bracketed at 0.03–0.08 $\mathrm{cm}^{-3}$, and a row at one tenth of the analogy density is printed beside the 47 Tucanae-analogy figures. Sgr A* was added as a consistency point (§3.5); the lowest-branch slopes were corrected to 0.59–0.71 (from 0.47–0.71) and the extrapolation depth to 2.9–3.7 decades at the median rate (from "three to five" and "four to six"). The set-fiducial distance of 5.49 kpc is stated, with the computations still at 5.43 kpc. The Strader et al. 0.2 $\mathrm{cm}^{-3}$ is labelled a gas number density, and outbound-request wording and one source-inconsistency item (the 3.3 $\mu$Jy label) were dropped from the data-availability section.

- **v2.2 (2026-10-08).** Gray-literature fixes: the chain behind the Mahida et al. gas density (Freire et al. M15 reading, Anderson upper limit), the Su et al. and Henleywillis et al. X-ray corroboration, Abbassi et al. (2026), the Tudor et al. footnote, and an extended provenance list.

- **v2.0 (R9-F-1).** (a) The efficiency law $\eta = 0.1\ \min(1, \dot{m}/10^{-2})$ was replaced by the Xie and Yuan (2012) piecewise fits with $\delta$ as an eighth nuisance, raising the natural-flow band by about two decades; the linear form corresponds to the last row of Table 6 (0.36, 0.47, 0.92 for the RIAF family). (b) The statistic reported as $P_{\mathrm{excl}}$, the fraction of natural draws exceeding a marginalized $\varepsilon_{95}$, was replaced by the posterior-predictive fraction of §3.6; the RIAF column moved from 19, 31 and 90 per cent to 78, 85 and 99 per cent at 6,000, 8,200 and $4 \times 10^4\ M_\odot$, and the abstract's statement that a natural flow survives comfortably at the fast-star anchor was withdrawn. (c) The infrared comparison was converted from band-peak to band-integrated luminosity, which loosened the RIAF anchors by 4 to 7 per cent, the jet anchors by 32 to 37 per cent and the radio-free anchors by 60 per cent; the infrared leg's effect on the RIAF limit fell from 2 to 0.4 per cent, the thin-disk X-ray and infrared legs, previously reported within 10 per cent of each other, separated by a factor of five, JWST ceased to be described as binding for the jet family, the mid-infrared forecast gain fell from 29, 31 and 21 per cent to 14, 17 and 9 per cent, and the limit-leg reconstruction effect fell from 29–30 per cent to 1 per cent. (d) The stellar-wind budget was recounted: $10^3$ giants,

$10^{-5}\,M_\odot\,\mathrm{yr}^{-1}$ and $n_e \sim 3\,\mathrm{cm}^{-3}$ became 8.4 giants, $8.4\times10^{-8}\,M_\odot\,\mathrm{yr}^{-1}$ and $0.030\,\mathrm{cm}^{-3}$. The budget had served as a floor ($n_e \geq 0.05\,\mathrm{cm}^{-3}$, claimed a factor 60 below the no-removal steady state and 4.6 below the 47 Tucanae value, and to return the widened prior almost to baseline); those claims were withdrawn. (e) The mass slope of the marginalized limit was corrected from the single-draw asymptote $M^{-3.3}$ to $M^{-2.1}$ to $M^{-2.2}$; the effect of the efficiency-prior floor from "about ten" to 9 and 40; and the deeper-radio forecast was reconciled with its realized factor of eleven. (f) Equation 10 was printed with a spurious second factor $[(\gamma+1)/2]^{-(\gamma+1)/2(\gamma-1)}$, which gave 0.286 and 0.362 as tabulated values and a $\gamma \to 5/3$ limit of $9/64 = 0.141$; no computed number used it. The Bondi-convention ratio of 1.9 and $\dot{M}_\mathrm{B} = 5.3\times10^{18}\,\mathrm{g\,s}^{-1}$ had been quoted as median values and belong to the 13th-percentile sound speed (1.43 and $3.5\times10^{18}$ at the median); the horizon-efficiency bracket $[10^{-4.1}, 10^{-2.5}]$ became $[10^{-4.4}, 10^{-2.6}]$, and the exchange factor to Haggard et al. (2013)'s convention 0.5–1.5 became 0.17–0.66. The rounded form of the Haggard efficiency span ($10^{-6}$–$10^{-8}$) had been quoted as the source's range. The revised code reproduces the v0.3 output to floating-point equality with these corrections disabled.

- **v1.x and earlier.** v0.2 restructured the analysis into a likelihood-based posterior with SED families, a gas-supply budget, a natural-expectation band and forecasts (referee memos of 2026-08-13). Before v0.2, a draft used $\mu_e = 1.5$ (tightening all limits by 27 per cent) and a common band-fraction bracket that left the JWST leg unconstraining. v0.3 corrected inputs: the X-ray limit was relabelled from $3\sigma$ (divided by 3) to the source's one-sided 95 per cent confidence (divided by 1.645), the Haggard reference distance became 5.2 kpc (an intermediate draft carried 4.8 kpc and a spurious 28 per cent shift in the X-ray leg; the leg is flux-based and unaffected), per-filter Chen limits replaced a single constant, and the duty-cycle epoch count rose from $N \sim 6$ (radio campaign as one epoch; sampling break $\delta \approx 0.1$) to $N \simeq 30$ ($\delta \approx 0.02$). v1.2 added the nuisance ladder, the band-adjudication figure, the continuous exclusion curve, the epoch-resolved duty-cycle analysis and the dispersion-measure requirement table. Appendix B retains the threshold-inversion construction that preceded the posterior.

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